

2.3.5 PROTOCOL DEFINITION

2.3.5.1 Sequence Rules

2.3.5.1.1 With the exception of abort primitives, only the sequence of primitives illustrated in figures 2.3.5-1 to 2.3.5-18 shall be permitted.

Note 1.— The following figures define the valid sequences of primitives that are possible to be invoked during the operation of the CPDLC application. It shows the relationship in time between the service request and the resulting indication, and if applicable, the subsequent response and resulting confirmation.

Note 2.— Abort primitives may interrupt and terminate any of the normal message sequences outlined below.

Note 3.— Primitives are processed in the order received.

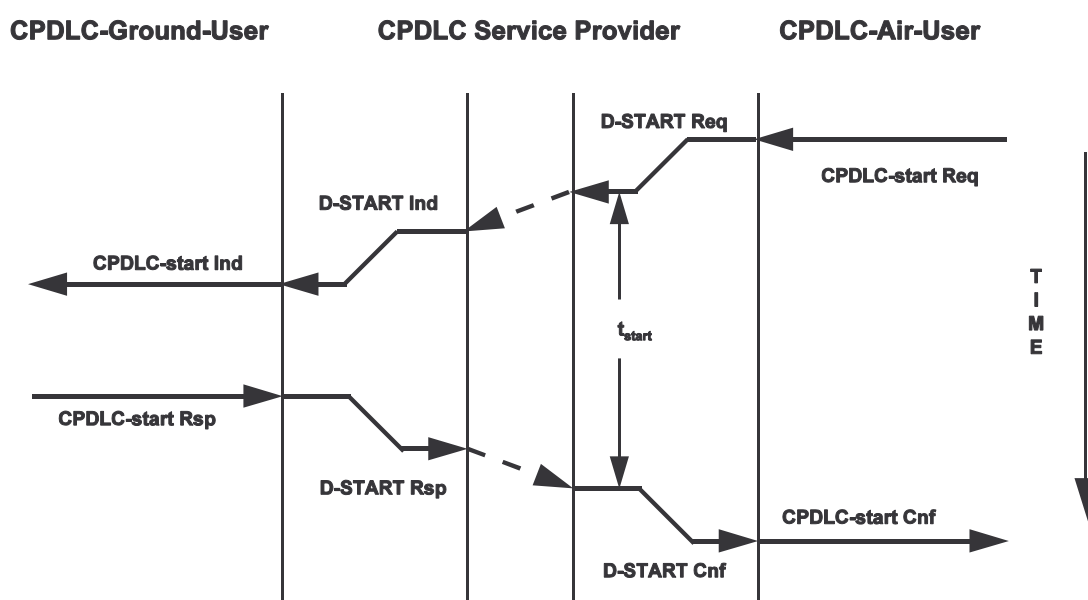


Figure 2.3.5-1. Sequence Diagram for CPDLC-start Service/Air Initiated

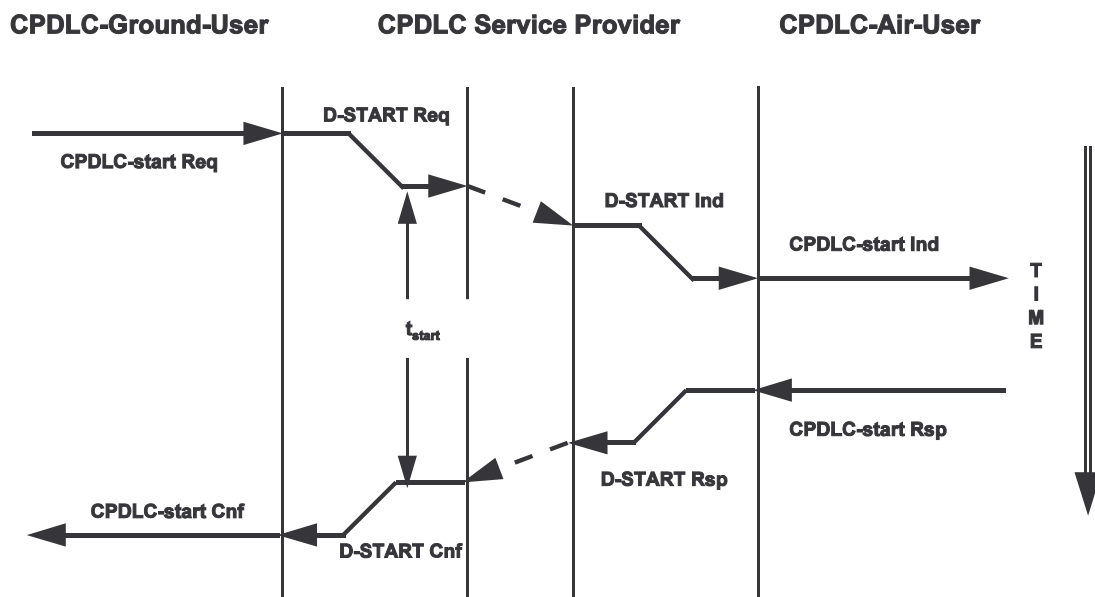


Figure 2.3.5-2. Sequence Diagram for CPDLC-start Service/Ground Initiated

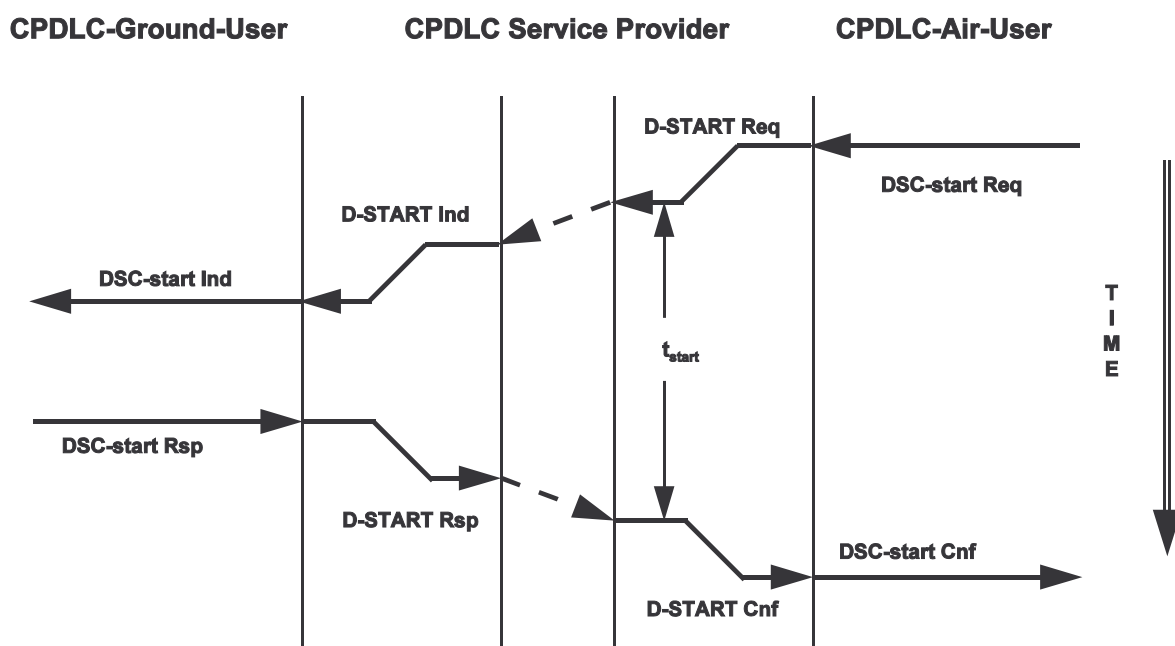


Figure 2.3.5-3. Sequence Diagram for DSC-start Service

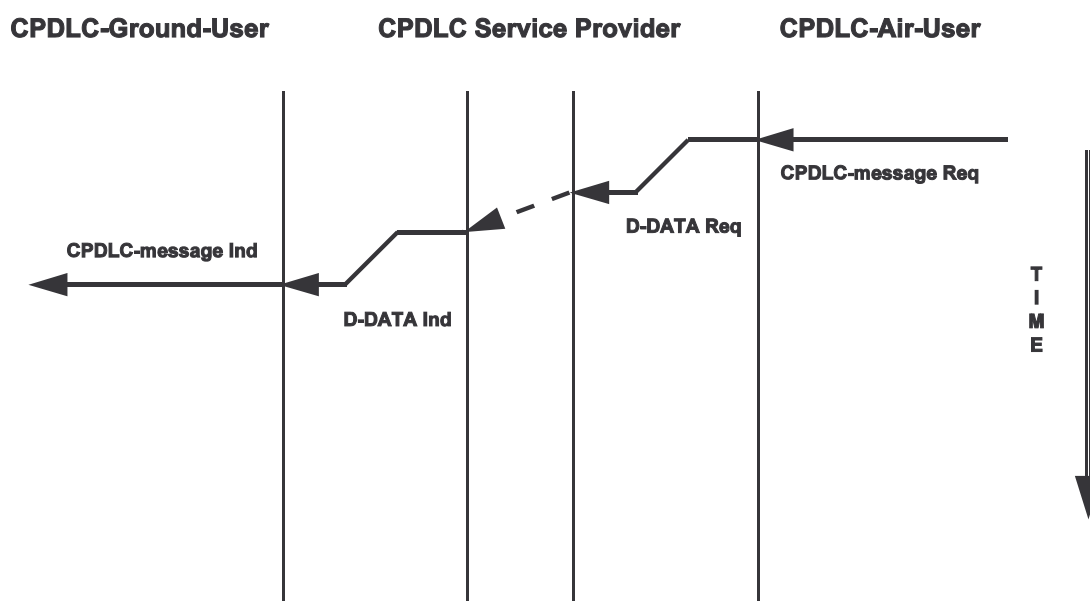


Figure 2.3.5-4. Sequence Diagram for CPDLC-message Service/Air Initiated

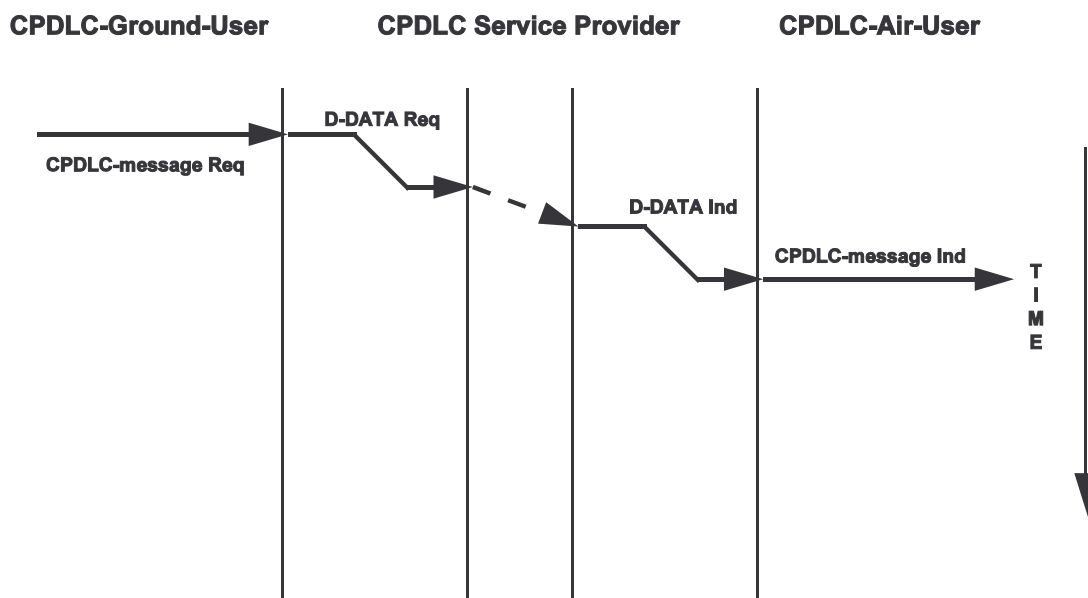


Figure 2.3.5-5. Sequence Diagram for CPDLC-message Service/Ground Initiated

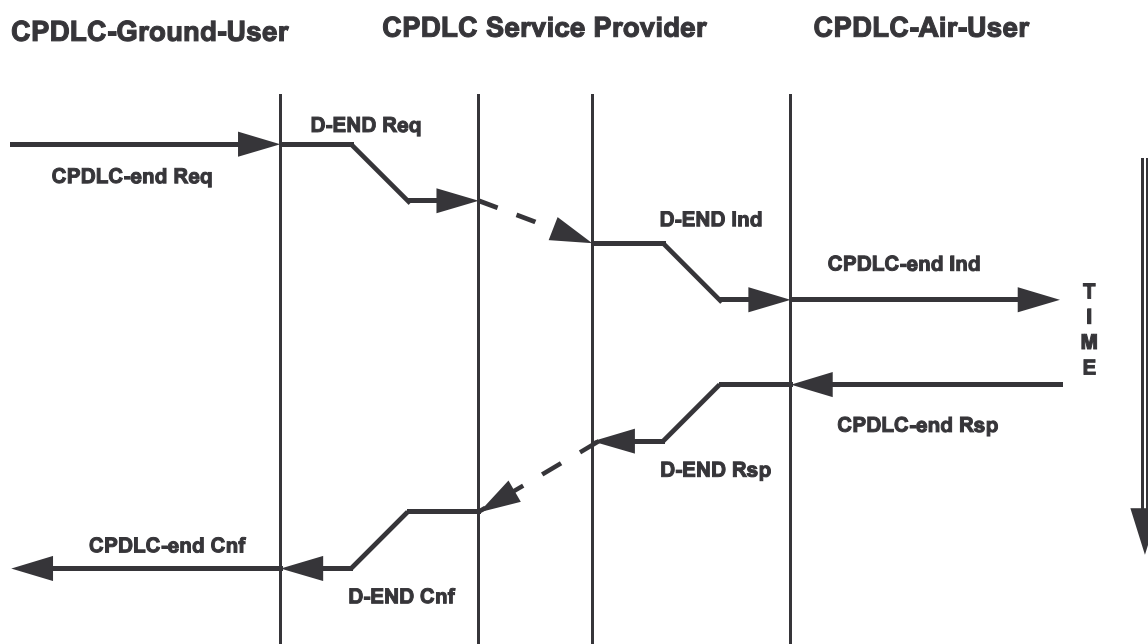


Figure 2.3.5-6. Sequence Diagram for CPDLC-end Service

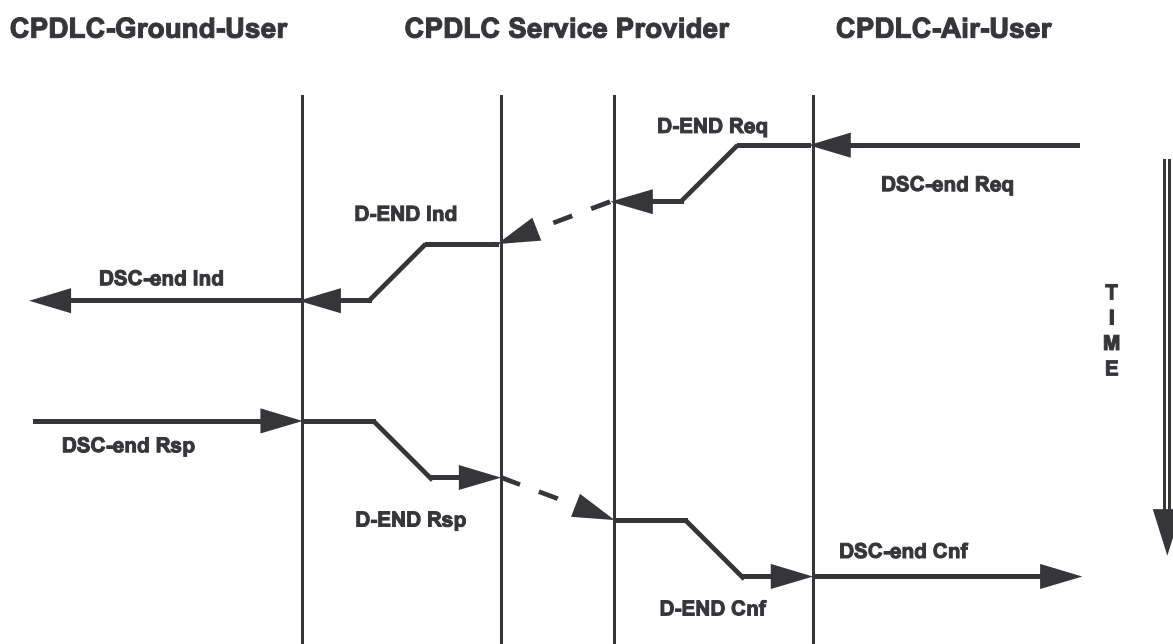


Figure 2.3.5-7. Sequence Diagram for DSC-end Service

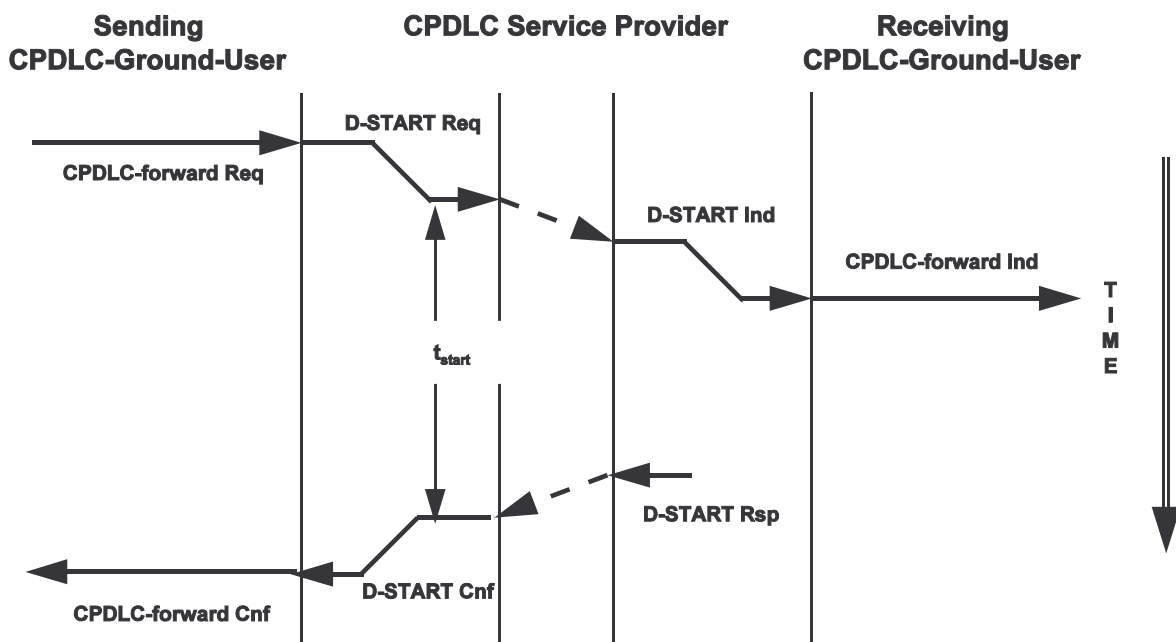


Figure 2.3.5-8. Sequence Diagram for CPDLC-forward Service/Ground Forwarding Supported, ASE Version Numbers the Same

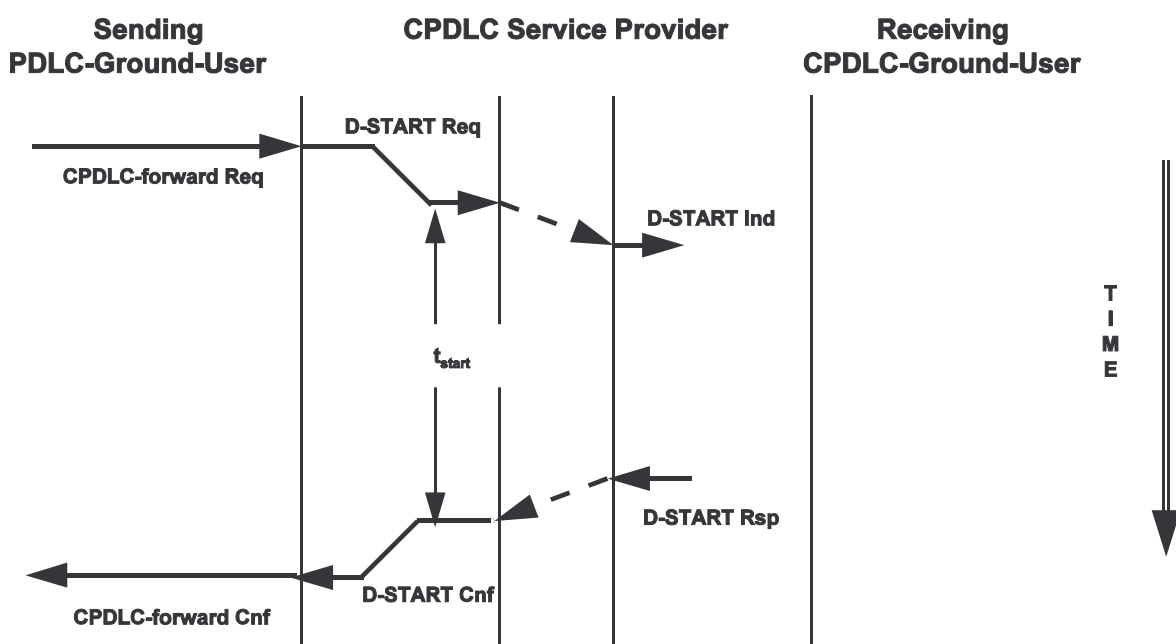


Figure 2.3.5-9. Sequence Diagram for CPDLC-forward Service/Ground Forwarding Not Supported, or Ground Forwarding Supported and ASE Version Numbers Not the Same

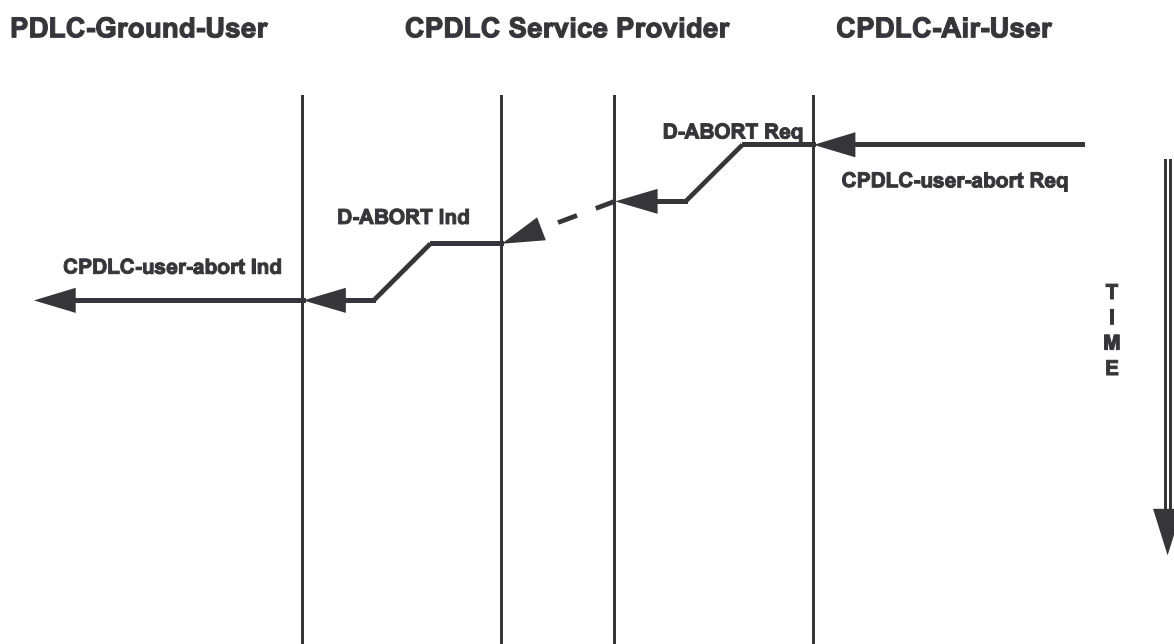


Figure 2.3.5-10. Sequence Diagram for CPDLC-user-abort Service/CPDLC-Air-User Initiated

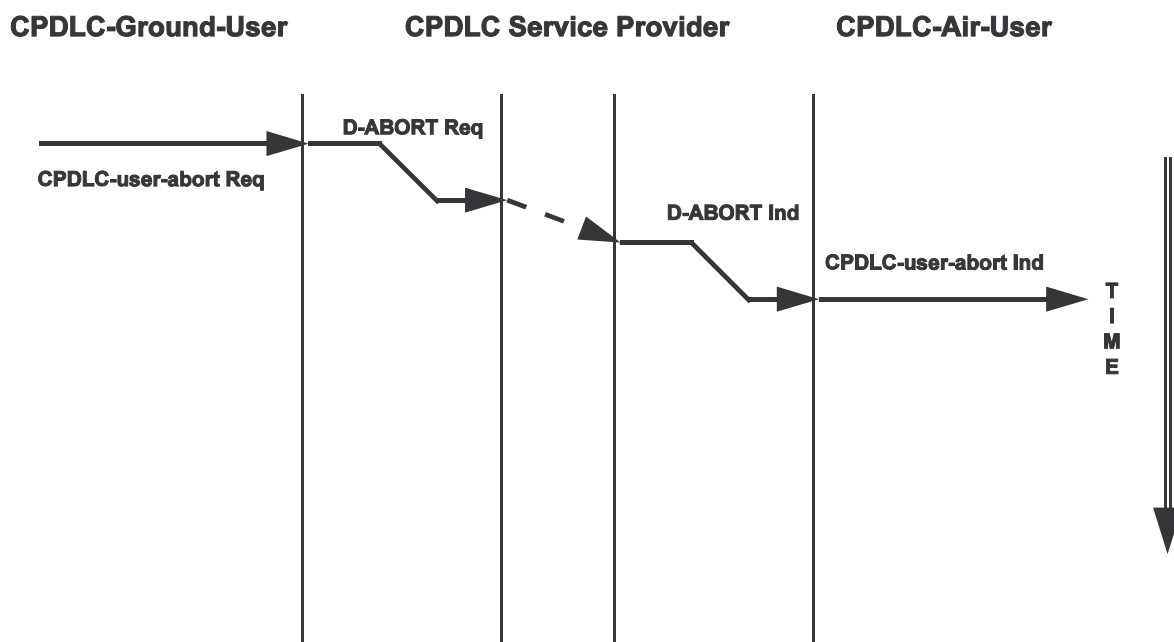


Figure 2.3.5-11. Sequence Diagram for CPDLC-user-abort Service/CPDLC-Ground-User Initiated

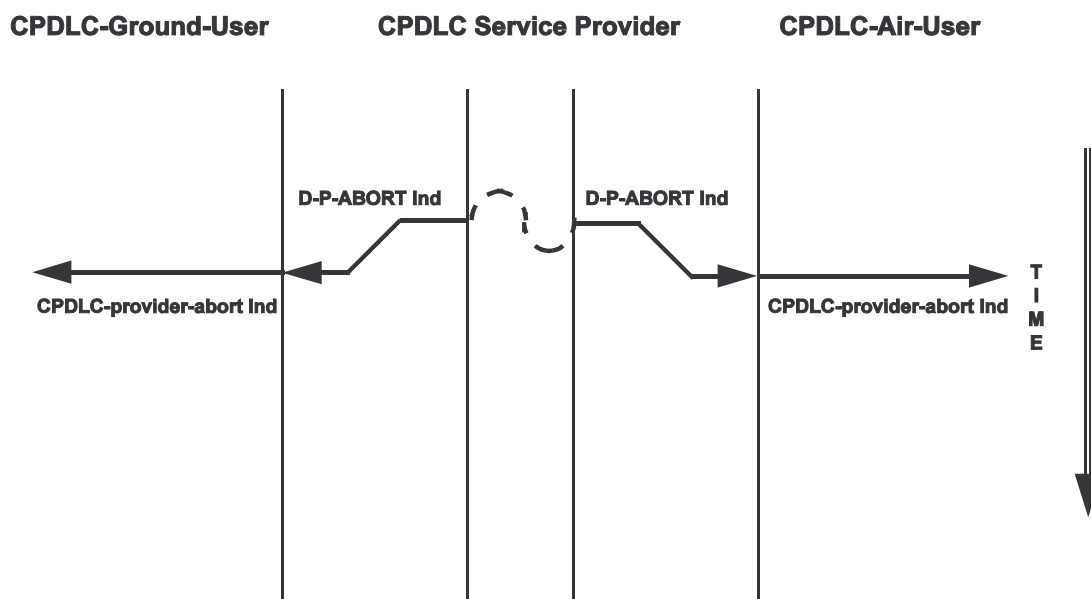


Figure 2.3.5-12. Sequence Diagram for CPDLC-provider-abort Service/Dialogue Service Abort

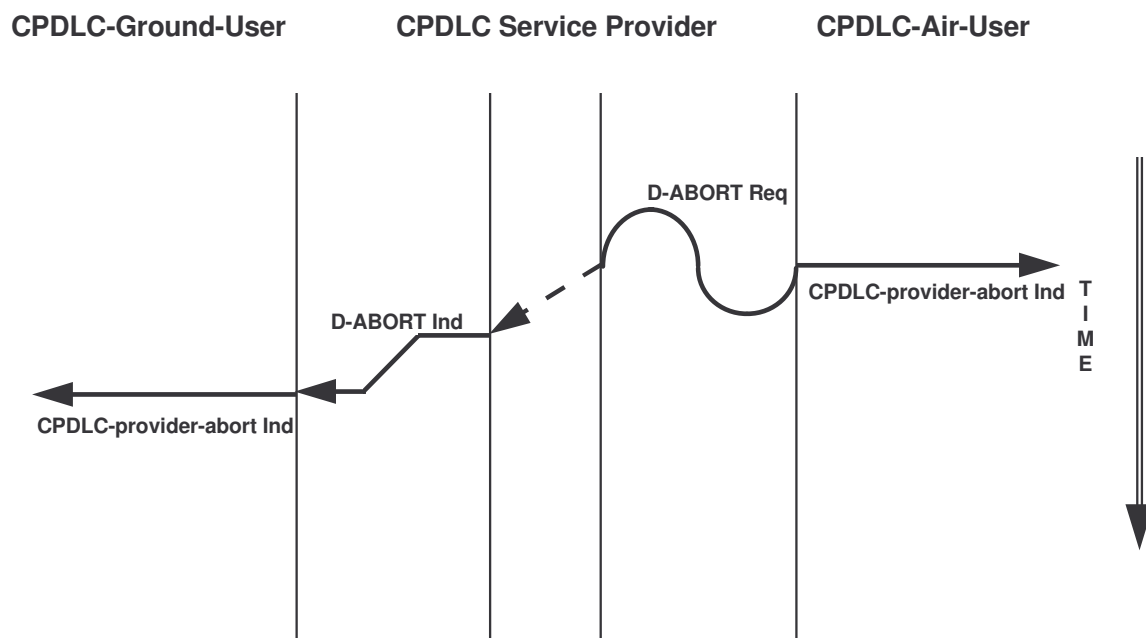


Figure 2.3.5-13. Sequence Diagram for CPDLC-provider-abort Service/CPDLC-Air-ASE Abort

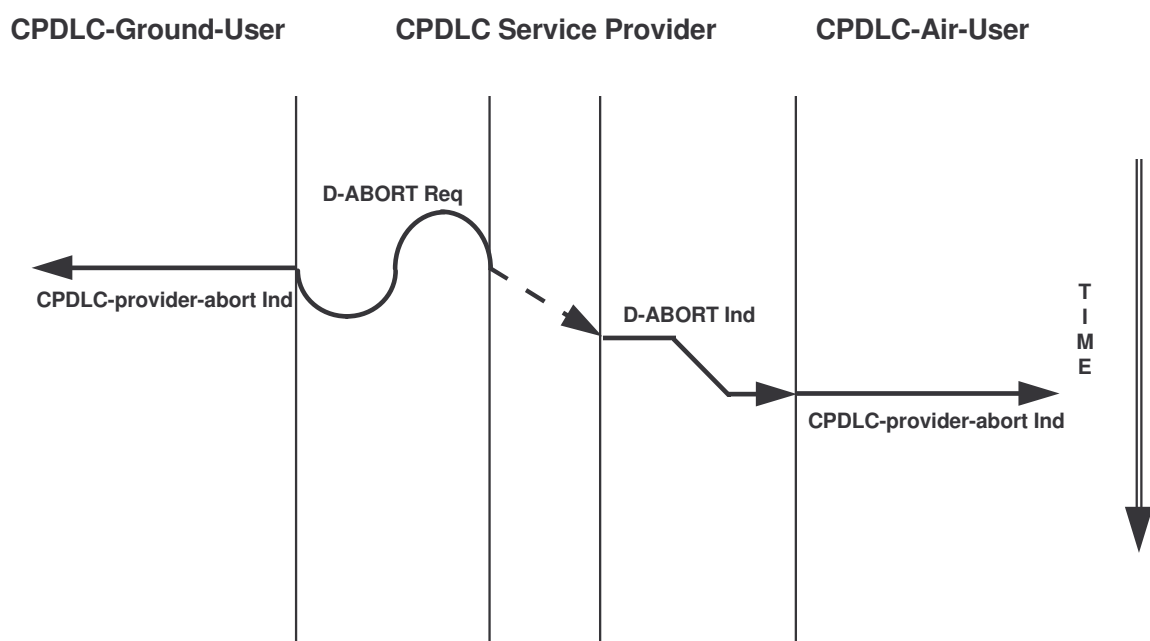


Figure 2.3.5-14. Sequence Diagram for CPDLC-provider-abort Service/
CPDLC-Ground-ASE Abort

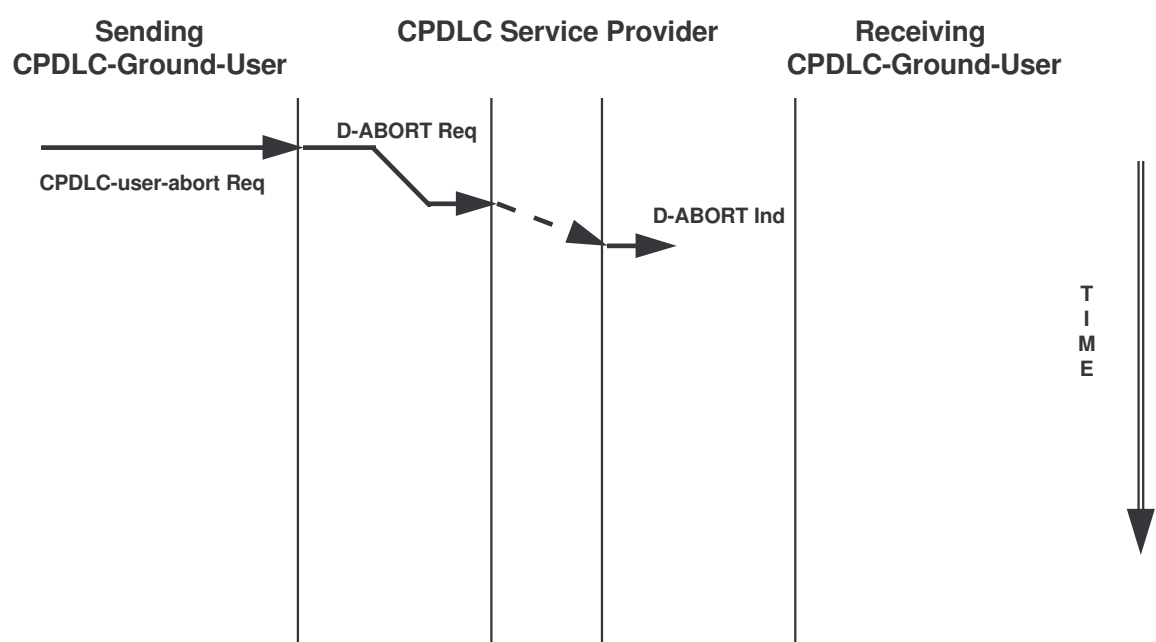


Figure 2.3.5-15. Sequence Diagram for CPDLC-user-abort Service/Sending
CPDLC-Ground-User Initiated

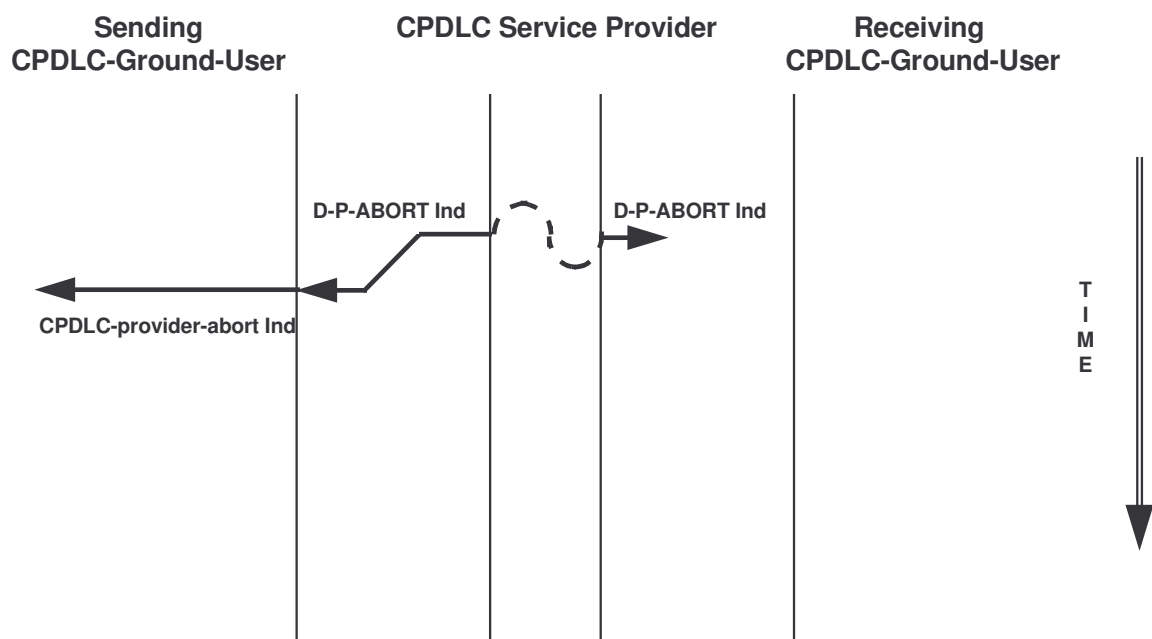


Figure 2.3.5-16. Sequence Diagram for CPDLC-provider-abort Service/Dialogue Service Abort

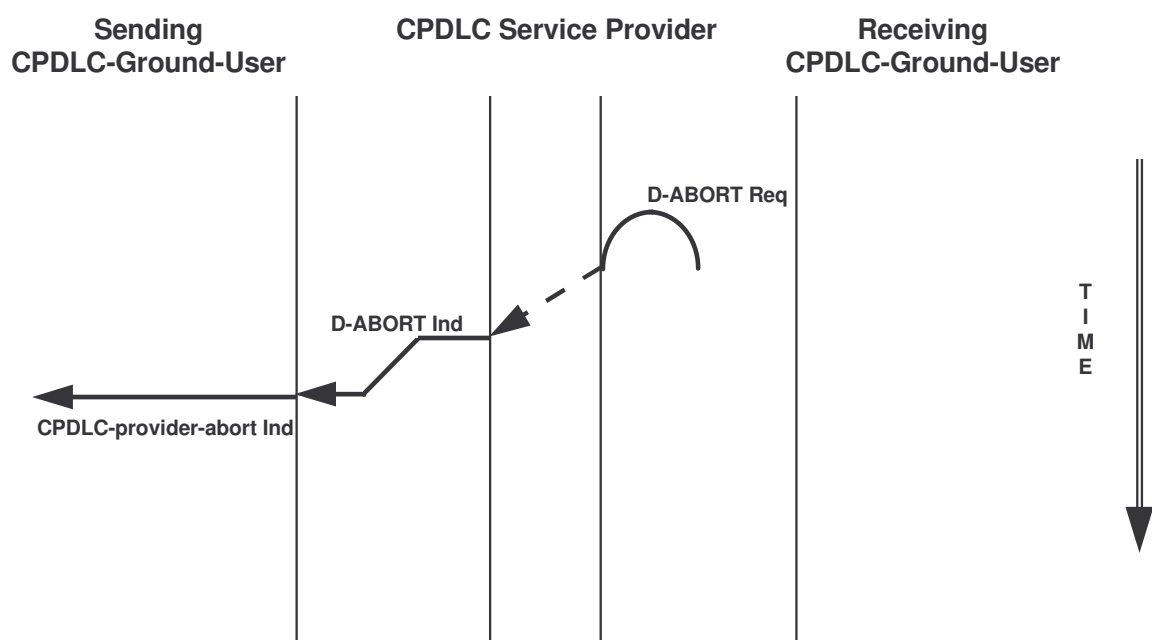


Figure 2.3.5-17. Sequence Diagram for CPDLC-provider-abort Service/Receiving CPDLC-Ground-ASE Abort

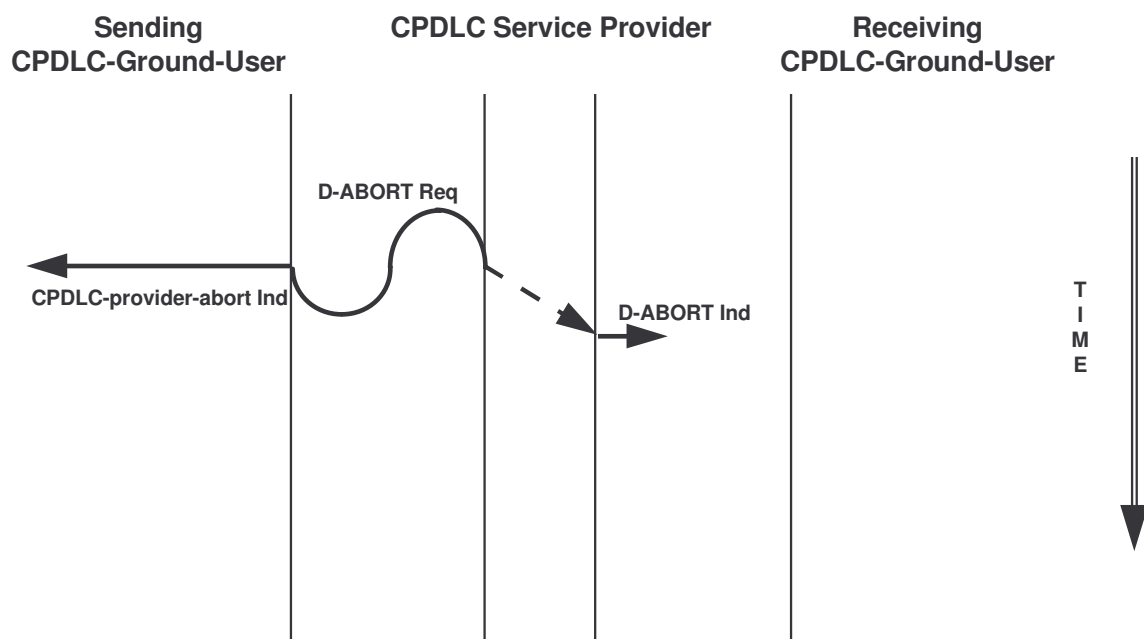


Figure 2.3.5-18. Sequence Diagram for CPDLC-provider-abort Service/Sending CPDLC-Ground-ASE Abort

2.3.5.2 CPDLC Service Provider Timers

2.3.5.2.1 A CPDLC-ASE shall be capable of detecting when a timer expires.

Note 1.— Table 2.3.5-1 lists the time constraints related to the CPDLC application. Each time constraint requires a timer to be set in the CPDLC protocol machine.

Note 2.— If the timer expires before the final event has occurred, a CPDLC-ASE takes appropriate action as defined in 2.3.5.4.1.

2.3.5.2.2 **Recommendation.** — *The timer values should be as indicated in Table 2.3.5-1.*

Table 2.3.5-1. CPDLC Service Provider Timers

CPDLC Service	Timer	Timer Value	Timer Start Event	Timer Stop Event
CPDLC-start	t_{start}	6 minutes	D-START request	D-START confirmation
DSC-start	t_{start}	6 minutes	D-START request	D-START confirmation
CPDLC-forward	t_{start}	6 minutes	D-START request	D-START confirmation

Note.— The receipt of CPDLC-user-abort requests, D-ABORT Indications, or D-P-ABORT Indications are also timer stop events.

2.3.5.3 CPDLC-air-ASE Protocol Description

2.3.5.3.1 Introduction

2.3.5.3.1.1 If no actions are described for a CPDLC service primitive when a CPDLC-air-ASE is in a specific state, then the invocation of that service primitive shall be prohibited while the CPDLC-air-ASE is in that state.

2.3.5.3.1.2 Upon receipt of a PDU, if no actions are described for the arrival of that PDU when a CPDLC-air-ASE is in a specific state, then that PDU is considered not permitted, and exception handling procedures as described in 2.3.5.4.4 shall apply.

2.3.5.3.1.3 If a PDU is received that cannot be decoded, then exception handling procedures as described in 2.3.5.4.3 for invalid PDU shall apply.

2.3.5.3.1.4 If a PDU is not received when one is required, then exception handling as described 2.3.5.4.3 shall apply.

Note 1.— The states defined for the CPDLC-air-ASE are the following.

- a) IDLE*
- b) START-REQ,*
- c) START-IND,*
- d) DIALOGUE, and*
- e) END.*

Note 2.— The CPDLC-air-user is an active user from:

- a) the time it has invoked the CPDLC-start service request until:*
 - 1) receipt of a CPDLC-start service confirmation with Result parameter equal to the abstract value “rejected”, or*
 - 2) invocation of a CPDLC-end service response with the Result parameter set to the abstract value “accepted”, or*
 - 3) invocation of a CPDLC-user-abort service request, or*
 - 4) receipt of CPDLC-user-abort service indication, or*
 - 5) receipt of a CPDLC-provider-abort service indication; or*
- b) the time it has received the CPDLC-start service indication until:*
 - 1) invocation of a CPDLC-start service response with Result parameter equal to the abstract value “rejected”, or*
 - 2) invocation of a CPDLC-end service response with the Result parameter set to the abstract value “accepted”, or*
 - 3) invocation of a CPDLC-user-abort service request, or*
 - 4) receipt of CPDLC-user-abort service indication, or*
 - 5) receipt of a CPDLC-provider-abort service indication; or*
- c) the time it has invoked the DSC-start service request until:*
 - 1) receipt of a DSC-start service confirmation with Result parameter equal to the abstract value “rejected”, or*

- 2) receipt of a DSC-end service confirmation with Result parameter equal to the abstract value “accepted”, or
- 3) invocation of a CPDLC-user-abort service request, or
- 4) receipt of CPDLC-user-abort service indication, or
- 5) receipt of a CPDLC-provider-abort service indication.

2.3.5.3.1.5 On initiation the CPDLC-air-ASE shall be in the *IDLE* state.

Note.— The CPDLC-air-ASE contains a Boolean called DSC. DSC has the abstract value “true” when the dialogue is a DSC dialogue, and has the abstract value “false” otherwise.

2.3.5.3.1.6 On the initiation of a CPDLC-air-ASE, DSC shall be set to the abstract value “false”.

2.3.5.3.2 D-START Indication

2.3.5.3.2.1 Upon receipt of a D-START indication, if the CPDLC-air-ASE is in the *IDLE* state and the D-START *User Data* parameter contains a GroundPDUs [ProtectedUplinkMessage(startup)] APDU, and the D-START *QOS Priority* parameter has the abstract value “high priority flight safety message” and the D-START *QOS Residual Error Rate* parameter has the abstract value “low”, the D-START *QOS Routing Class* parameter has one of the abstract values specified in Table 2.3.6-1, and the D-START *Calling Peer ID* parameter is a valid four to eight character facility designation and the D-START *Security Requirements* parameter is consistent with the local security policy (as defined in Annex 10 Volume III Part I Chapter 3), the CPDLC-air-ASE shall:

- a) Invoke CPDLC-start service indication containing the following:
 - 1) the D-START *Calling Peer ID* parameter value as the CPDLC-start service *Calling Peer Identifier* parameter value,
 - 2) the D-START *QOS Routing Class* parameter value as the CPDLC-start service *Class of Communication* parameter value,
 - 3) the GroundPDUs APDU-element as the CPDLC-start service *Protected CPDLC Message* parameter value, and
- b) Enter the *START-IND* state.

2.3.5.3.3 D-START Confirmation

2.3.5.3.3.1 Upon receipt of a D-START confirmation, if the CPDLC-air-ASE is in the *START-REQ* state and the D-START *Result* parameter has the abstract value “accepted” and DSC has the abstract value “false” and D-START *User Data* parameter contains a GroundPDUs [ProtectedUplinkMessage(send)] APDU and the D-START *Security Requirements* parameter value is the same as the one set for the D-START request, the CPDLC-air-ASE shall:

- a) Stop timer t_{start} .
- b) Invoke CPDLC-start service confirmation containing the following:
 - 1) the APDU contained in the D-START *User Data* parameter as the CPDLC-start service *Response Message* parameter value, and
 - 2) the abstract value “accepted” as the CPDLC-start service *Result* parameter value, and
- c) Enter the *DIALOGUE* state.

2.3.5.3.3.2 Upon receipt of a D-START confirmation, if the CPDLC-air-ASE is in the *START-REQ* state and the D-START *Result* parameter has the abstract value “rejected (permanent)” and the D-START *Reject Source* parameter has the abstract value “DS user” and DSC has the abstract value “false” and the D-START *User Data* parameter contains a GroundPDUs [ProtectedUplinkMessage(send)] APDU and the D-START *Security Requirements* parameter value is the same as the one set for the D-START request, the CPDLC-air-ASE shall:

- a) Stop timer t_{start} .
- b) Invoke CPDLC-start service confirmation containing the following:
 - 1) the APDU contained in the D-START *User Data* parameter as the CPDLC-start service *Response Message* parameter value, and
 - 2) the abstract value “rejected” as the CPDLC-start service *Result* parameter value, and
- c) Enter the *IDLE* state.

2.3.5.3.3.3 Upon receipt of a D-START confirmation, if the CPDLC-air-ASE is in the *START-REQ* state and the D-START *Result* parameter has the abstract value “accepted” and DSC has the abstract value “true” and D-START *User Data* parameter contains a GroundPDUs [ProtectedUplinkMessage(send)] APDU and the D-START *Security Requirements* parameter value is the same as the one set for the D-START request, the CPDLC-air-ASE shall:

- a) Stop timer t_{start} .
- b) Invoke DSC-start service confirmation containing the following:
 - 1) the APDU contained in the D-START *User Data* parameter as the DSC-start service *Response Message* parameter value, and
 - 2) the abstract value “accepted” as the DSC-start service *Result* parameter value, and
- c) Enter the *DIALOGUE* state.

2.3.5.3.3.4 Upon receipt of a D-START confirmation, if the CPDLC-air-ASE is in the *START-REQ* state and the D-START *Result* parameter has the abstract value “rejected (permanent)” and the D-START *Reject Source* parameter has the abstract value “DS user”, and DSC has the abstract value “true”, and the D-START *User Data* parameter contains a GroundPDUs [ProtectedUplinkMessage(send)] APDU, the CPDLC-air-ASE shall:

- a) Stop timer t_{start}
- b) Invoke DSC-start service confirmation containing the following:
 - 1) the APDU contained in the D-START *User Data* parameter as the DSC-start service *Response Message* parameter value, and
 - 2) the abstract value “rejected” as the DSC-start service *Result* parameter value,
- c) Set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.3.4 D-DATA Indication

2.3.5.3.4.1 Upon receipt of a D-DATA indication, if the CPDLC-air-ASE is in the *DIALOGUE* state and the APDU contained in the D-DATA *User Data* parameter is a GroundPDUs [ProtectedUplinkMessage(send)] APDU, the CPDLC-air-ASE shall:

- a) Invoke CPDLC-message service indication with the APDU contained in the D-DATA *User Data* parameter as the CPDLC-message service *Protected CPDLC Message* parameter value, and
- b) Remain in the *DIALOGUE* state.

2.3.5.3.4.2 Upon receipt of a D-DATA indication, if the CPDLC-air-ASE is in the *END* state and DSC has the abstract value of “true” and the APDU contained in the D-DATA *User Data* parameter is a GroundPDUs [ProtectedUplinkMessage(send)] APDU, the CPDLC-air-ASE shall:

- a) Invoke CPDLC-message service indication with the APDU contained in the D-DATA *User Data* parameter as the CPDLC-message service *Protected CPDLC Message* parameter value, and
- b) Remain in the *END* state.

2.3.5.3.5 D-END Indication

2.3.5.3.5.1 Upon receipt of a D-END indication, if the CPDLC-air-ASE is in the *DIALOGUE* state, and DSC has the abstract value “false”, and the D-END *User Data* parameter contains a GroundPDUs [ProtectedUplinkMessage(send)] APDU, the CPDLC-air-ASE shall:

- a) Invoke CPDLC-end service indication with the APDU contained in the D-END *User Data* parameter as the CPDLC-end service *Protected* CPDLC Message parameter value, and
- b) Enter the *END* state.

2.3.5.3.6 D-END Confirmation

2.3.5.3.6.1 Upon receipt of a D-END confirmation, if the CPDLC-air-ASE is in the *END* state and the abstract the D-END *Result* parameter has the abstract value “accepted” and DSC has the abstract value “true” and the D-END *User Data* parameter contains a GroundPDUs [ProtectedUplinkMessage(send)] APDU, the CPDLC-air-ASE shall:

- a) Invoke DSC-end service confirmation with:
 - 1) the APDU contained in the D-END *User Data* parameter as the DSC-end service *Protected* CPDLC Message parameter value, and
 - 2) the abstract value “accepted” as the CPDLC-end service *Result* parameter value,
- b) Set DSC to the abstract value “false”, and
- c) Enter the *IDLE* state.

2.3.5.3.6.2 Upon receipt of a D-END confirmation, if the CPDLC-air-ASE is in the *END* state and the D-END *Result* parameter has the abstract value “rejected”, and DSC has the abstract value “true”, and the D-END *User Data* parameter contains a GroundPDUs [ProtectedUplinkMessage(send)] APDU, the CPDLC-air-ASE shall:

- a) Invoke DSC-end service confirmation with:
 - 1) the APDU contained in the D-END *User Data* parameter as the DSC-end service *Protected* CPDLC Message parameter value, and
 - 2) the abstract value “rejected” as the CPDLC-end service *Result* parameter value,
- b) Enter the *DIALOGUE* state.

2.3.5.3.7 CPDLC-start Service Request

2.3.5.3.7.1 Upon receipt of a CPDLC-start service request, if the CPDLC-air-ASE is in the *IDLE* state, the CPDLC-air-ASE shall:

- a) Create an AircraftPDUs APDU with a StartDownMessage APDU element containing:
 - 1) the abstract value “cpdlc” as the mode,
 - 2) the *Protected* CPDLC Message parameter as the ProtectedDownlinkMessage,
- b) Invoke D-START request with the following:
 - 1) the CPDLC-start service *Called Peer Identifier* parameter value as the D-START *Called Peer ID* parameter value,
 - 2) the CPDLC-start service *Calling Peer Identifier* parameter value as the D-START *Calling Peer ID* parameter value,
 - 3) the CPDLC-start service *Security Required* parameter value if specified by the CPDLC-user else the abstract value “no security” as the D-START *Security Requirements* parameter value,
 - 4) the D-START *Quality of Service* parameters set as follows:
 - i) if provided, the CPDLC-start service *Class of Communication* parameter value as the D-START *QOS Routing Class* parameter value, or
 - ii) The abstract value of “high priority flight safety messages”, as the D-START *QOS Priority* parameter value, and
 - iii) The abstract value of “low” as the D-START *QOS Residual Error Rate* parameter value, and
 - 5) the APDU as the D-START *User Data* parameter value;
- c) Start timer t_{start} and
- d) Enter the *START-REQ* state.

2.3.5.3.8 CPDLC-start Service Response

2.3.5.3.8.1 Upon receipt of a CPDLC-start service response, if the CPDLC-air-ASE is in the *START-IND* state and the CPDLC-start service *Result* parameter has the abstract value “accepted”, and DSC has the abstract value “false”, the CPDLC-air-ASE shall:

- a) Create an AircraftPDUs APDU with a ProtectedDownlinkMessage APDU element based on the *Response Message* parameter,
- b) Invoke D-START response with the following:
 - 1) the APDU as the D-START *User Data* parameter,
 - 2) the abstract value received in the D-START indication *Security Requirements* parameter as the D-START *Security Requirements* parameter,
 - 3) the abstract value “accepted” as the D-START *Result* parameter value, and
- c) Enter the *DIALOGUE* state

2.3.5.3.8.2 Upon receipt of a CPDLC-start service response, if the CPDLC-air-ASE is in the *START-IND* state, and the CPDLC-start service *Result* parameter has the abstract value “rejected” and DSC has the abstract value “false”, the CPDLC-air-ASE shall:

- a) Create an AircraftPDUs APDU with a ProtectedDownlinkMessage APDU element based on the *Response Message* parameter,
- b) Invoke D-START response with the following:
 - 1) the APDU as the D-START *User Data* parameter,
 - 2) the abstract value received in the D-START indication *Security Requirements* parameter as the D-START *Security Requirements* parameter,
 - 3) the abstract value “rejected (permanent)” as the D-START *Result* parameter value, and
- c) Enter the *IDLE* state.

2.3.5.3.9 DSC-start Service Request

2.3.5.3.9.1 Upon receipt of a DSC-start service request, if the CPDLC-air-ASE is in the *IDLE* state, the CPDLC-air-ASE shall:

- a) Create an AircraftPDUs APDU with a StartDownMessage APDU element containing:
 - 1) the abstract value “dsc” as the mode, and
 - 2) the *Protected* CPDLC Message parameter as the ProtectedDownlinkMessage,
- b) Invoke D-START request with the following:

- 1) the DSC-start service *Facility Designation* parameter value as the D-START *Called Peer ID* parameter value,
- 2) the DSC-start service *Aircraft Address* parameter value as the D-START *Calling Peer ID* parameter value,
- 3) the DSC-start service *Security Required* parameter value if specified by the CPDLC-air-user else the abstract value “no security” as the D-START *Security Requirements* parameter value,
- 4) Set the D-START *Quality of Service* parameters as follows:
 - i) if provided, the DSC-START service *Class of Communication* parameter value as the D-START *QOS Routing Class* parameter value,
 - ii) The abstract value of “high priority flight safety messages”, as the D-START *QOS Priority* parameter value, and
 - iii) The abstract value of “low” as the D-START *QOS Residual Error Rate* parameter value, and
- 5) the APDU as the D-START *User Data* parameter value;
- c) Set DSC to the abstract value “true”,
- d) Start timer t_{start} and
- e) Enter the *START-REQ* state.

2.3.5.3.10 CPDLC-message Service Request

2.3.5.3.10.1 Upon receipt of a CPDLC-message service request and the *Protected CPDLC Message* parameter contains a CPDLC Message, if the CPDLC-air-ASE is in the *DIALOGUE* state, the CPDLC-air-ASE shall:

- a) Create an AircraftPDUs APDU with a ProtectedDownlinkMessage APDU-element based on the CPDLC-message service *Protected CPDLC Message* parameter,
- b) Invoke D-DATA request with the APDU as the D-DATA *User Data* parameter value, and
- c) Remain in the *DIALOGUE* state.

2.3.5.3.10.2 Upon receipt of a CPDLC-message service request and the *Protected CPDLC Message* parameter contains a CPDLC Message, if the CPDLC-air-ASE is in the *END* state and DSC has the abstract value “false”, the CPDLC-air-ASE shall:

- a) Create an AircraftPDUs APDU with a ProtectedDownlinkMessage APDU-element based on the CPDLC-message service *Protected* CPDLC Message parameter,
- b) Invoke D-DATA request with the APDU as the D-DATA *User Data* parameter value, and
- c) Remain in the *END* state.

2.3.5.3.11 CPDLC-end Service Response

2.3.5.3.11.1 Upon receipt of a CPDLC-end service response, if the CPDLC-air-ASE is in the *END* state, and the CPDLC-end service *Result* parameter has the abstract value “accepted” and DSC has the abstract value “false”, the CPDLC-air-ASE shall:

- a) Create an AircraftPDUs APDU with a ProtectedDownlinkMessage APDU-element based on the CPDLC-end service *Protected* CPDLC Message parameter,
- b) Invoke D-END response with the following:
 - 1) the APDU as the D-END *User Data* parameter value, and
 - 2) the abstract value “accepted”, as the D-END *Result* parameter value, and
- c) Enter the *IDLE* state.

2.3.5.3.11.2 Upon receipt of a CPDLC-end service response, if the CPDLC-air-ASE is in the *END* state, and the CPDLC-end service *Result* parameter has the abstract value “rejected” and DSC has the abstract value “false”, the CPDLC-air-ASE shall:

- a) Create an AircraftPDUs APDU with a ProtectedDownlinkMessage APDU-element based on the CPDLC-end service *Protected* CPDLC Message parameter,
- b) Invoke D-END response with the following:
 - 1) the APDU as the D-END *User Data* parameter value, and
 - 2) the abstract value “rejected”, as the D-END *Result* parameter value, and
- c) Enter the *DIALOGUE* state.

2.3.5.3.12 DSC-end Service Request

2.3.5.3.12.1 Upon receipt of a DSC-end service request, if the CPDLC-air-ASE is in the *DIALOGUE* state and DSC has the abstract value “true”, the CPDLC-air-ASE shall:

- a) Create an AircraftPDUs APDU with a ProtectedDownlinkMessage APDU-element based on the DSC-end service *Protected* CPDLC Message parameter,

- b) Invoke D-END request with the APDU as the D-END *User Data* parameter value, and
- c) Enter the *END* state.

2.3.5.3.13 CPDLC-user-abort Service Request

2.3.5.3.13.1 Upon receipt of a CPDLC-user-abort service request, if the CPDLC-air-ASE is not in the *IDLE* state, the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-user-abort service *Reason* parameter is provided, create an AircraftPDUs APDU with a CPDLCUserAbortReason APDU-element based on the CPDLC-user-abort service *Reason* parameter,
- c) Else create an AircraftPDUs APDU with a CPDLCUserAbortReason [undefined] APDU-element,
- d) Invoke D-ABORT request with the following:
 - 1) the D-ABORT *Originator* parameter set to the abstract value “user”, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.3.14 D-ABORT Indication

2.3.5.3.14.1 Upon receipt of a D-ABORT indication, if the CPDLC-air-ASE is not in the *IDLE* state, and the D-ABORT *Originator* parameter is “user” and the D-ABORT *User Data* parameter contains a GroundPDUs [CPDLCUserAbortReason] APDU, the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-air-user is an active user, invoke CPDLC-user-abort service indication with the APDU contained in the D-ABORT *User Data* parameter as the CPDLC-user-abort service *Reason* parameter value,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.3.14.2 Upon receipt of a D-ABORT indication, if the CPDLC-air-ASE is not in the *IDLE* state, and if the D-ABORT *Originator* parameter is “provider” and if the D-ABORT *User Data* parameter is provided, the D-ABORT *User Data* parameter contains a GroundPDUs [CPDLCProviderAbortReason] APDU, the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the D-ABORT *User Data* parameter as the CPDLC-provider-abort service *Reason* parameter value, if provided,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.3.15 D-P-ABORT Indication

2.3.5.3.15.1 Upon receipt of a D-P-ABORT indication, if the CPDLC-air-ASE is not in the *IDLE* state, the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the CPDLC-provider-abort service *Reason* parameter set to the abstract value “communication-service-failure”,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.4 CPDLC-air-ASE Exception Handling

2.3.5.4.1 A Timer Expires

2.3.5.4.1.1 If a CPDLC-air-ASE detects that a timer has expired, that CPDLC-air-ASE shall:

- a) Interrupt any current activity,
- b) Create an AircraftPDUs APDU with a CPDLCProviderAbortReason [timer-expired] APDU message element,
- c) Invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and

- d) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the abstract value “timer-expired” as the CPDLC-provider abort service *Reason* parameter value,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.4.2 Unrecoverable System Error

2.3.5.4.2.1 **Recommendation.** — *If a CPDLC-air-ASE has an unrecoverable system error, the CPDLC-air-ASE should:*

- a) *Stop any timer,*
- b) *Create an AircraftPDUs APDU with a CPDLCProviderAbortReason [undefined-error] APDU message element,*
- c) *Invoke D-ABORT request with:*
 - 1) *the abstract value “provider” as the D-ABORT Originator parameter value, and*
 - 2) *the APDU as the D-ABORT User Data parameter value, and*
- d) *If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the abstract value “undefined-error” as the CPDLC-provider abort service Reason parameter value,*
- e) *If DSC has the abstract value “true”, set DSC to the abstract value “false”, and*
- f) *Enter the IDLE state.*

2.3.5.4.3 Invalid PDU

2.3.5.4.3.1 If the *User Data* parameter of a D-END confirmation with *Result* parameter set to the abstract value “rejected”, or the *User Data* parameter of a D-START confirmation with *Result* parameter set to the abstract value “accepted”, or if the *User Data* parameter of a D-START indication, a D-DATA indication, or a D-END indication, does not contain a valid PDU, the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) Create an AircraftPDUs APDU with a CPDLCProviderAbortReason [invalid-PDU] APDU message element,

- c) Invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and
- d) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the abstract value “invalid-PDU” as the CPDLC-provider abort service *Reason* parameter value,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.4.3.2 If the *User Data* parameter of a D-START confirmation with *Result* set to the abstract value “rejected (permanent)”, or a D-END confirmation with *Result* set to the abstract value “accepted”, is not a valid PDU then the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the CPDLC-provider-abort service *Reason* parameter set to the abstract value “invalid-PDU” ,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.4.4 Protocol Error

2.3.5.4.4.1 If the *User Data* parameter of a D-START indication, D-START confirmation, D-DATA indication, or D-END indication is a valid PDU, but is not a PDU for which action is described within a given state in 2.3.5.3, the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) Create an AircraftPDUs APDU with a CPDLCProviderAbortReason [protocol-error] APDU message element,
- c) Invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and

- d) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the abstract value “protocol-error” as the CPDLC-provider abort service *Reason* parameter value ,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.4.4.2 (deleted)

2.3.5.4.4.3 If the *User Data* parameter of a D-END confirmation is a valid PDU, but is not a permitted PDU as defined in 2.3.5.3, the CPDLC-air-ASE shall:

- a) If the D-END *Result* parameter is set to the abstract value “rejected”, then
 - 1) Stop any timer,
 - 2) Create an AircraftPDUs APDU with a CPDLCProviderAbortReason [protocol-error] APDU message element,
 - 3) Invoke D-ABORT request with:
 - iv) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - v) the APDU as the D-ABORT *User Data* parameter value, and
- b) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the abstract value “protocol-error” as the CPDLC-provider-abort service *Reason* parameter value,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.4.4.4 Upon receipt of a Dialogue service primitive for which there are no instruction in 2.3.5.3 (i.e. the primitive was not expected or was expected under other conditions or with other parameter values), the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) Create an AircraftPDUs APDU with a CPDLCProviderAbortReason [protocol-error] APDU message element,
- c) If a dialogue exists, invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and

- 2) the APDU as the D-ABORT *User Data* parameter value, and
- d) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the abstract value “protocol-error” as the CPDLC-provider abort service *Reason* parameter value ,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.4.5 D-START Confirmation *Result* or *Reject Source* Not as Expected

2.3.5.4.5.1 If a D-START confirmation *Result* parameter has the abstract value of “rejected (transient)” or if the *Reject Source* parameter has the abstract value of “DS provider”, the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the CPDLC-provider-abort service *Reason* parameter set to the abstract value “communication-service-error”,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.4.6 D-START Indication *Quality of Service* Not as Expected

2.3.5.4.6.1 If a D-START indication *QOS Priority* parameter does not have the abstract value of “high priority flight safety messages” or if the *QOS Residual Error Rate* parameter does not have the abstract value of “low”, or if the *QOS Routing Class* parameter does not have one of the abstract values specified in Table 2.3.6-1, the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) Create an AircraftPDUs APDU with a CPDLCProviderAbortReason [invalid-QOS-parameter] APDU message element,
- c) Invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as D-ABORT *User Data* parameter value,
- d) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- e) Enter the *IDLE* state.

2.3.5.4.7 Expected PDU Missing

2.3.5.4.7.1 If the *User Data* parameter of a D-START indication or confirmation, a D-DATA indication, or a D-END indication or confirmation does not contain a PDU, the CPDLC-air-ASE shall:

- a) Stop any timer,
- b) Create an AircraftPDUs APDU with a CPDLCProviderAbortReason [expected-PDU-missing] APDU message element,
- c) Invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as the D-ABORT *User Data* parameter values,
- d) If the CPDLC-air-user is an active user, invoke CPDLC-provider-abort service indication with the abstract value “not-permitted-PDU” as the CPDLC-provider-abort service *Reason* parameter value,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.4.8 D-START Security Requirements Parameter Not as Expected

Note.— This section applies to the case when the D-START Security Requirements parameter does not meet the local security policy (as defined in Annex 10 Volume III Part I Chapter 3).

2.3.5.4.8.1 Upon receipt of a D-START indication with the *Security Requirements* parameter not consistent with the local security policy (as defined in Annex 10 Volume III Part I Chapter 3) or upon receipt of a D-START confirmation with the *Security Requirements* parameter not equal to the value that was set in the D-START request, the CPDLC-air-ASE shall:

- a) Stop all timers,
- b) If the dialogue with the peer is still open:
 - 1) Create an AircraftPDUs APDU with a CPDLCProviderAbortReason [communication-service-failure] APDU message element,
 - 2) Invoke D-ABORT request with:
 - i) the abstract value “provider” as the D-ABORT *Originator* parameter value, and

- ii) the APDU as the D-ABORT *User Data* parameter value,
- c) If the CPDLC-air-user is active, invoke CPDLC-provider-abort service indication with the abstract value “communication-service-failure” as the CDPLC-provider-abort service *Reason* parameter value,
- d) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- e) Enter the *IDLE* state.

2.3.5.5 CPDLC-ground-ASE Protocol Description

2.3.5.5.1 Introduction

2.3.5.5.1.1 If no actions are described for a CPDLC service primitive when a CPDLC-ground-ASE is in specific state, then the invocation of that service primitive shall be prohibited while the CPDLC-ground-ASE is in that state.

2.3.5.5.1.2 Upon receipt of a PDU, if no actions are described for the arrival of that PDU when a CPDLC-ground-ASE is in a specific state, then that PDU is considered not permitted, and exception handling procedures as described in 2.3.5.6.4 shall apply.

2.3.5.5.1.3 If a PDU is received that cannot be decoded, then exception handling procedures as described in 2.3.5.6.3 for invalid PDU shall apply.

2.3.5.5.1.4 If a PDU is not received when one is required, then exception handling as described in 2.3.5.6.3 shall apply.

Note 1.— The states defined for the CPDLC-ground-ASE are the following.

- a) *IDLE*
- b) *START-REQ,*
- c) *START-IND,*
- d) *DIALOGUE,*
- e) *END, and*
- f) *FORWARD.*

Note 2.— The CPDLC-ground-user is an active user from:

- a) *the time it has invoked the CPDLC-start service request until:*

- 1) receipt of a CPDLC-start service confirmation with Result parameter equal to the abstract value “rejected”, or
 - 2) receipt of a CPDLC-end service confirmation with the Result parameter equal to the abstract value “accepted”, or
 - 3) invocation of a CPDLC--user-abort service request, or
 - 4) receipt of a CPDLC-user-abort service indication, or
 - 5) receipt of a CPDLC-provider-abort service indication; or
- b) the time it has received the CPDLC-start service indication until:
- 1) invocation of a CPDLC-start service response with Result parameter set to the abstract value “rejected”, or
 - 2) receipt of a CPDLC-end service confirmation with the Result parameter equal to the abstract value “accepted”, or
 - 3) invocation of a CPDLC-user-abort service request, or
 - 4) receipt of CPDLC-user-abort service indication, or
 - 5) receipt of a CPDLC-provider-abort service indication; or
- c) the time it has received the DSC-start service indication until:
- 1) invocation of a DSC-start service response with Result parameter equal to the abstract value “rejected”, or
 - 2) invocation of a CPDLC-user-abort service request, or
 - 3) receipt of CPDLC-user-abort service indication, or
 - 4) receipt of a CPDLC-provider-abort service indication; or
- d) the time it has invoked the CPDLC-forward service request until:
- 1) receipt of a CPDLC-forward service confirmation,
 - 2) invocation of a CPDLC-user-abort service request, or
 - 3) receipt of CPDLC-user-abort service indication, or
 - 4) receipt of a CPDLC-provider-abort service indication.

2.3.5.5.1.5 On initiation the CPDLC-ground-ASE shall be in the *IDLE* state.

Note.— The CPDLC-ground-ASE contains a Boolean called *DSC*. *DSC* has the abstract value “true” when the dialogue is a DSC dialogue, and has the abstract value “false” otherwise.

2.3.5.5.1.6 On the initiation of a CPDLC-ground-ASE, *DSC* shall be set to the abstract value “false”.

2.3.5.5.2 D-START Indication

2.3.5.5.2.1 Upon receipt of a D-START indication, if the CPDLC-ground-ASE is in the *IDLE* state, and the abstract value of the D-START *Calling Peer ID* parameter is a 24-bit aircraft address, and the D-START *User Data* parameter contains an AircraftPDUs [StartDownMessage] APDU with the APDU-element mode “cpdlc”, and the D-START *QOS Priority* parameter has the abstract value “high priority flight safety messages” and the D-START *QOS Residual Error Rate* parameter has the abstract value “low” and the D-START *QOS Routing Class* parameter has one of the abstract values specified in Table 2.3.6-1 and the D-START *Security Requirements* parameter is consistent with the local security policy (as defined in Annex 10 Volume III Part I Chapter 3), the CPDLC-ground-ASE shall:

- a) Invoke CPDLC-start service indication containing the following:
 - 1) the D-START *Calling Peer ID* parameter value as the CPDLC-start service *Calling Peer Identifier* parameter value,
 - 2) the D-START *QOS Routing Class* parameter value as the CPDLC-start service *Class of Communication* parameter value, and
 - 3) the AircraftPDUs APDU-element as the CPDLC-start service *Protected CPDLC Message* parameter value, and
- b) Enter the *START-IND* state.

2.3.5.5.2.2 Upon receipt of a D-START indication, if the CPDLC-ground-ASE is in the *IDLE* state, and the abstract value of the D-START *Calling Peer ID* parameter is a 24-bit aircraft address, and the D-START *User Data* parameter contains an AircraftPDUs [StartDownMessage]APDU with the APDU-element mode “dsc”, and the D-START *QOS Priority* parameter has the abstract value “high priority flight safety messages” and the D-START *QOS Residual Error Rate* parameter has the abstract value “low”, and the D-START *QOS Routing Class* parameter has one of the abstract values specified in Table 2.3.6-1, and the D-START *Security Requirements* parameter is consistent with the local security policy (as defined in Annex 10 Volume III Part I Chapter 3), the CPDLC-ground-ASE shall:

- a) Invoke DSC-start service indication containing the following:
 - 1) the D-START *Calling Peer ID* parameter value as the DSC-start service *Aircraft Address* parameter value,
 - 2) the D-START *QOS Routing Class* parameter value as the CPDLC-start service *Class of Communication* parameter value,
 - 3) the AircraftPDUs APDU as the DSC-start service *Protected CPDLC Message* parameter value,

- b) Set DSC to “true”, and
- c) Enter the *START-IND* state.

2.3.5.5.2.3 Upon receipt of a D-START indication, if the CPDLC-ground-ASE is in the *IDLE* state, and the abstract value of the D-START *Calling Peer ID* parameter is a Facility Designation, and the D-START *User Data* parameter contains a GroundPDUs [ATCForwardMessage] APDU and the CPDLC-ground-ASE supports the CPDLC-forward service, and the D-START *QOS Priority* parameter has the abstract value “high priority flight safety messages” and the D-START *QOS Residual Error Rate* parameter has the abstract value “low”, the CPDLC-ground-ASE shall:

- a) If the D-START *DS User Version Number* parameter value is equal to the CPDLC-ground-ASE version number and the D-START *Security Requirements* parameter is consistent with the local security policy (as defined in Annex 10 Volume III Part I Chapter 3):
 - 1) Invoke CPDLC-forward service indication containing the following:
 - i) the D-START *Calling Peer ID* parameter value as the CPDLC-forward service *Calling Facility Designation* parameter value,
 - ii) set the D-START GroundPDUs APDU-element as the CPDLC-forward service *CPDLC Message* parameter value, and
 - 2) Create a GroundPDUs APDU with an ATCForwardResponse [success] APDU element,
 - 3) Invoke D-START response with the following:
 - i) the APDU as the D-START *User Data* parameter value, and
 - ii) the abstract value “rejected (permanent)” as the D-START *Result* parameter value, and
 - 4) Remain in the *IDLE* state.
- b) If the D-START *DS User Version Number* parameter value is not equal to the CPDLC-ground-ASE version number:
 - 1) Create a GroundPDUs APDU with an ATCForwardResponse [version-not-equal] APDU element,
 - 2) Invoke D-START response with the following:
 - i) the CPDLC-ground-ASE version number as the D-START *DS User Version Number* parameter value,

- ii) the APDU as the D-START *User Data* parameter value, and
 - iii) the abstract value “rejected (permanent)” as the D-START *Result* parameter value, and
- 3) Remain in the *IDLE* state.

2.3.5.5.2.4 Upon receipt of a D-START indication, if the CPDLC-ground-ASE is in the *IDLE* state, and the abstract value of the D-START *Calling Peer ID* parameter is a Facility Designation, and the D-START *User Data* parameter contains a GroundPDUs APDU and the APDU element is an ATCForwardMessage and the CPDLC-ground-ASE does not support the CPDLC-forward service and the D-START *Security Requirements* parameter is consistent with the local security policy (as defined in Annex 10 Volume III Part I Chapter 3), the CPDLC-ground-ASE shall:

- a) Create a GroundPDUs APDU with an ATCForwardResponse [service-not-supported] APDU element,
- b) Invoke D-START response with the following:
 - 1) the APDU as the D-START *User Data* parameter value, and
 - 2) the abstract value “rejected (permanent)” as the D-START *Result* parameter value, and
- c) Remain in the *IDLE* state.

2.3.5.5.3 D-START Confirmation

2.3.5.5.3.1 Upon receipt of a D-START confirmation, if the CPDLC-ground-ASE is in the *START-REQ* state and if the D-START *Result* parameter has the abstract value “accepted”, and DSC has the abstract value of “false” and D-START *User Data* parameter contains an AircraftPDUs [ProtectedDownlinkMessage] APDU and the D-START *Security Requirements* parameter value is the same as the one set for the D-START request, the CPDLC-ground-ASE shall:

- a) Stop timer t_{start} ,
- b) Invoke CPDLC-start service confirmation containing the following:
 - 1) the APDU contained in the D-START *User Data* parameter as the CPDLC-start service *Response Message* parameter value, and
 - 2) the abstract value “accepted” as the CPDLC-start service *Result* parameter value, and
- c) Enter the *DIALOGUE* state.

2.3.5.5.3.2 Upon receipt of a D-START confirmation, if the CPDLC-ground-ASE is in the *START-REQ* state and the D-START *Result* parameter has the abstract value “rejected (permanent)” and the D-START *Reject Source* parameter has the abstract value “DS user” and DSC has the abstract value “false” and the D-START *User Data* parameter contains an AircraftPDUs [ProtectedDownlinkMessage] APDU and the D-START *Security Requirements* parameter value is the same as the one set for the D-START request, the CPDLC-ground-ASE shall:

- a) Stop timer t_{start}
- b) Invoke CPDLC-start service confirmation containing the following:
 - 1) the APDU contained in the D-START *User Data* parameter as the CPDLC-start service Response Message parameter value, and
 - 2) the abstract value “rejected” as the CPDLC-start service *Result* parameter value, and
- c) Enter the *IDLE* state.

2.3.5.5.3.3 Upon receipt of a D-START confirmation, if the CPDLC-ground-ASE is in the *FORWARD* state and if the D-START *Result* parameter has the abstract value “rejected (permanent)” and the *Reject Source* parameter has the abstract value “DS user” and the D-START *User Data* parameter contains a GroundPDUs [ATCForwardResponse] APDU and the D-START *Security Requirements* parameter value is the same as the one set for the D-START request, the CPDLC-ground-ASE shall:

- a) Stop timer t_{start}
- b) invoke CPDLC-forward service confirmation with:
 - 1) the D-START GroundPDUs APDU element as the CPDLC-forward service *Result* parameter value, and
 - 2) if provided, the D-START *DS User Version Number* parameter value as the CPDLC-forward service *ASE Version Number* parameter value, and
- c) enter the *IDLE* state.

2.3.5.5.4 D-DATA Indication

2.3.5.5.4.1 Upon receipt of a D-DATA indication, if the CPDLC-ground-ASE is in the *DIALOGUE* state and the APDU contained in the D-DATA *User Data* parameter is a AircraftPDUs [ProtectedDownlinkMessage] APDU, the CPDLC-ground-ASE shall:

- a) Invoke CPDLC-message service indication with the APDU contained in the D-DATA *User Data* parameter as the CPDLC-message service *Protected CPDLC Message* parameter value, and
- b) Remain in the *DIALOGUE* state.

2.3.5.5.4.2 Upon receipt of a D-DATA indication, if the CPDLC-ground-ASE is in the *END* state and DSC has the abstract value “false” and the APDU contained in the D-DATA *User Data* parameter is an AircraftPDUs [ProtectedDownlinkMessage] APDU, the CPDLC-ground-ASE shall:

- a) Invoke CPDLC-message service indication with the APDU contained in the D-DATA *User Data* parameter as the CPDLC-message service *Protected CPDLC Message* parameter value, and
- b) Remain in the *END* state.

2.3.5.5.5 D-END Indication

2.3.5.5.5.1 Upon receipt of a D-END indication, if the CPDLC-ground-ASE is in the *DIALOGUE* state, and DSC has the abstract value “true”, and the D-END *User Data* parameter contains an AircraftPDUs [ProtectedDownlinkMessage] APDU, the CPDLC-ground-ASE shall:

- a) Invoke DSC-end service indication with the APDU contained in the D-END *User Data* parameter as the DSC-end service *Protected CPDLC Message* parameter value, and
- b) Enter the *END* state.

2.3.5.5.6 D-END Confirmation

2.3.5.5.6.1 Upon receipt of a D-END confirmation, if the CPDLC-ground-ASE is in the *END* state and the D-END *Result* parameter has the abstract value “accepted” and DSC has the abstract value “false” and the D-END *User Data* parameter contains an AircraftPDUs [ProtectedDownlinkMessage] APDU, the CPDLC-ground-ASE shall:

- a) Invoke CPDLC-end service confirmation with:
 - 1) The APDU contained in the D-END *User Data* parameter as the CPDLC-end service *Protected CPDLC Message* parameter value, and
 - 2) The abstract value “accepted” as the CPDLC-end service *Result* parameter value, and
- b) Enter the *IDLE* state.

2.3.5.5.6.2 Upon receipt of a D-END confirmation, if the CPDLC-ground-ASE is in the *END* state and the D-END *Result* has the abstract value “rejected” and DSC has the abstract value “false”, and the D-END *User Data* parameter contains an AircraftPDUs [ProtectedDownlinkMessage] APDU, the CPDLC-ground-ASE shall:

- a) Invoke CPDLC-end service confirmation with:

- 1) The APDU contained in the D-END *User Data* parameter as the CPDLC-end service *Protected* CPDLC Message parameter value, and
 - 2) The abstract value “rejected” as the CPDLC-end service *Result* parameter value, and
- b) Enter the *DIALOGUE* state.

2.3.5.5.7 CPDLC-start Service Request

2.3.5.5.7.1 Upon receipt of a CPDLC-start service request, if the CPDLC-ground-ASE is in the *IDLE* state, the CPDLC-ground-ASE shall:

- a) Create a GroundPDUs APDU with an UplinkMessage(startup) APDU element containing the *Protected* CPDLC Message parameter as the UplinkMessage,
- b) Invoke D-START request with the following:
 - 1) the CPDLC-start service *Called Peer Identifier* parameter value as the D-START *Called Peer ID* parameter value,
 - 2) the CPDLC-start service *Calling Peer Identifier* parameter value as the D-START *Calling Peer ID* parameter value,
 - 3) the CPDLC-start service *Security Required* parameter value if specified by the CPDLC-user else the abstract value “no security” as the D-START *Security Requirements* parameter value,
 - 4) the D-START *Quality of Service* parameters set as follows:
 - i) if provided, the CPDLC-start service *Class of Communication* parameter value as the D-START *QOS Routing Class* parameter value,
 - ii) The abstract value of “high priority flight safety messages”, as the D-START *QOS Priority* parameter value, and
 - iii) The abstract value of “low” as the D-START *QOS Residual Error Rate* parameter value, and
 - 5) the APDU as the D-START *User Data* parameter value;
- c) Start timer t_{start} and
- d) Enter the *START-REQ* state.

2.3.5.5.8 CPDLC-start Service Response

2.3.5.5.8.1 Upon receipt of a CPDLC-start service response, if the CPDLC-ground-ASE is in the *START-IND* state and the CPDLC-start service *Result* parameter has the abstract value “accepted”, and DSC has the abstract value “false”, the CPDLC-ground-ASE shall:

- a) create a GroundPDUs APDU with a ProtectedUplinkMessage(send) APDU element based on the *Response Message* parameter,
- b) Invoke D-START response with the following:
 - 1) The APDU as the D-START *User Data* parameter value,
 - 2) the abstract value received in the D-START indication *Security Requirements* parameter as the D-START *Security Requirements* parameter,
 - 3) the abstract value “accepted” as the D-START *Result* parameter value, and
- c) Enter the *DIALOGUE* state.

2.3.5.5.8.2 Upon receipt of a CPDLC-start service response, if the CPDLC-ground-ASE is in the *START-IND* state and the CPDLC-start service *Result* parameter has the abstract value “rejected” and DSC has the abstract value “false”, the CPDLC-ground-ASE shall:

- a) create a GroundPDUs APDU with a ProtectedUplinkMessage(send) APDU element based on the *Response Message* parameter,
- b) Invoke D-START response with the following:
 - 1) The APDU as the D-START *User Data* parameter value,
 - 2) the abstract value received in the D-START indication *Security Requirements* parameter as the D-START *Security Requirements* parameter, and
 - 3) the abstract value “rejected (permanent)” as the D-START *Result* parameter value, and
- c) Enter the *IDLE* state.

2.3.5.5.9 DSC-start Service Response

2.3.5.5.9.1 Upon receipt of a DSC-start service response, if the CPDLC-ground-ASE is in the *START-IND* state and the DSC-start service *Result* parameter has the abstract value “accepted” and DSC has the abstract value “true”, the CPDLC-ground-ASE shall:

- a) create a GroundPDUs APDU with a ProtectedUplinkMessage(send) APDU element based on the *Response Message* parameter,

- b) Invoke D-START response with the following:
 - 1) The APDU element as D-START *User Data* parameter value, and
 - 2) the abstract value received in the D-START indication *Security Requirements* parameter as the D-START *Security Requirements* parameter, and
 - 3) the abstract value “accepted” as the D-START *Result* parameter value, and
- c) Enter the *DIALOGUE* state.

2.3.5.5.9.2 Upon receipt of a DSC-start service response, if the CPDLC-ground-ASE is in the *START-IND* state and the DSC-start service *Result* parameter has the abstract value “rejected” and DSC has the abstract value “true”, the CPDLC-ground-ASE shall:

- a) create a GroundPDUs APDU with a ProtectedUplinkMessage(send) APDU element based on the *Response Message* parameter,
- b) Invoke D-START response with the following:
 - 1) The APDU element as D-START *User Data* parameter value, and
 - 2) the abstract value received in the D-START indication *Security Requirements* parameter as the D-START *Security Requirements* parameter, and
 - 3) the abstract value “rejected (permanent)” as the D-START *Result* parameter value,
- c) Set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.5.10 CPDLC-message Service Request

2.3.5.5.10.1 Upon receipt of a CPDLC-message service request and the *Protected CPDLC Message* parameter contains a CPDLC Message, if the CPDLC-ground-ASE is in the *DIALOGUE* state, the CPDLC-ground-ASE shall:

- a) Create a GroundPDUs APDU with a ProtectedUplinkMessage(send) APDU element based on the CPDLC-message service *Protected CPDLC Message* parameter,
- b) Invoke D-DATA request with the APDU as the D-DATA *User Data* parameter value, and
- c) Remain in the *DIALOGUE* state.

2.3.5.5.10.2 Upon receipt of a CPDLC-message service request and the *Protected CPDLC Message* parameter contains a CPDLC Message, if the CPDLC-ground-ASE is in the *END* state and DSC has the abstract value “true”, the CPDLC-ground-ASE shall:

- a) Create a GroundPDUs APDU with a ProtectedUplinkMessage(send) APDU element based on the CPDLC-message service *Protected CPDLC Message* parameter,
- b) Invoke D-DATA request with the APDU as the D-DATA *User Data* parameter value, and
- c) Remain in the *END* state.

2.3.5.5.11 CPDLC-end Service Request

2.3.5.5.11.1 Upon receipt of a CPDLC-end service request, if the CPDLC-ground-ASE is in the *DIALOGUE* state and DSC has the abstract value “false”, the CPDLC-ground-ASE shall:

- a) Create a GroundPDUs APDU with a ProtectedUplinkMessage(send) APDU element based on the CPDLC-end service *Protected CPDLC Message* parameter,
- b) Invoke D-END request with the APDU as the D-END *User Data* parameter value, and
- c) Enter the *END* state.

2.3.5.5.12 DSC-end Service Response

2.3.5.5.12.1 Upon receipt of a DSC-end service response, if the CPDLC-ground-ASE is in the *END* state and DSC has the abstract value “true”, and the DSC-end service *Result* parameter has the abstract value “accepted”, the CPDLC-ground-ASE shall:

- a) Create a GroundPDUs APDU with a ProtectedUplinkMessage(send) APDU element based on the DSC-end service *Protected CPDLC Message* parameter,
- b) Invoke D-END response with the following:
 - 1) the APDU as the D-END *User Data* parameter; and
 - 2) the abstract value “accepted” as the D-END *Result* parameter value,
- c) Set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.5.12.2 Upon receipt of a DSC-end service response, if the CPDLC-ground-ASE is in the *END* state and DSC has the abstract value “true”, and the DSC-end service *Result* parameter has the abstract value “rejected”, the CPDLC-ground-ASE shall:

- a) Create a GroundPDUs APDU with a ProtectedUplinkMessage(send) APDU element based on the DSC-end service *Protected CPDLC Message* parameter,
- b) Invoke D-END response with the following:
 - 1) the APDU as the D-END *User Data* parameter; and
 - 2) the abstract value “rejected” as the D-END *Result* parameter value, and
- c) Enter the *DIALOGUE* state.

2.3.5.5.13 CPDLC-forward Service Request

2.3.5.5.13.1 Upon receipt of a CPDLC-forward service request, if the CPDLC-ground-ASE is in the *IDLE* state, the CPDLC-ground-ASE shall:

- a) Create a GroundPDUs APDU with an ATCForwardMessage APDU element based on the CPDLC-forward service *CPDLC Message* parameter,
- b) Invoke D-START request with the following:
 - 1) the CPDLC-forward service *Called Facility Designation* parameter value as the D-START *Called Peer ID* parameter value,
 - 2) the CPDLC-start service *Calling Facility Designation* parameter value as the D-START *Calling Peer ID* parameter value,
 - 3) the CPDLC-forward service *Security Required* parameter value if specified by the CPDLC-user else the abstract value “no security” as the D-START *Security Requirements* parameter value,
 - 4) the D-START *Quality of Service* parameters set as follows:
 - i) if provided, the CPDLC-forward service *Class of Communication* parameter value as the D-START *QOS Routing Class* parameter value,
 - ii) The abstract value of “high priority flight safety messages”, as the D-START *QOS Priority* parameter value, and
 - iii) The abstract value of “low” as the D-START *QOS Residual Error Rate* parameter value, and
 - 5) the APDU as the D-START *User Data* parameter value;
- c) Start timer t_{start} , and

- d) Enter the *FORWARD* state.

2.3.5.5.14 CPDLC-user-abort Service Request

2.3.5.5.14.1 Upon receipt of a CPDLC-user-abort service request, if the CPDLC-ground-ASE is not in the *IDLE* state, the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-user-abort service *Reason* parameter is provided, create a GroundPDUs APDU with a CPDLCUserAbortReason APDU element based on the CPDLC-user-abort service *Reason* parameter,
- c) Else create a GroundPDUs APDU with a CPDLCUserAbortReason [undefined] APDU element,
- d) Invoke D-ABORT request with the following:
 - 1) the D-ABORT *Originator* parameter set to the abstract value “user”, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.5.15 D-ABORT Indication

2.3.5.5.15.1 Upon receipt of a D-ABORT indication, if the CPDLC-ground-ASE is not in the *IDLE* state and the D-ABORT *Originator* parameter is “user” and the D-ABORT *User Data* parameter contains an AircraftPDUs [CPDLCUserAbortReason] APDU, the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-Ground-User is an active user, invoke CPDLC-user-abort service indication with the APDU contained in the D-ABORT *User Data* parameter as the CPDLC-user-abort service *Reason* parameter value,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.5.15.2 Upon receipt of a D-ABORT indication, if the CPDLC-ground-ASE is not in the *IDLE* state, and if the D-ABORT *Originator* parameter is “provider” and if the D-ABORT *User Data* parameter is provided, the D-ABORT *User Data* parameter contains either an AircraftPDUs [CPDLCProviderAbortReason] APDU or a GroundPDUs [CPDLCProviderAbortReason] APDU, the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the D-ABORT *User Data* parameter as the CPDLC-provider-abort service *Reason* parameter value, if provided,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.5.16 D-P-ABORT Indication

2.3.5.5.16.1 Upon receipt of a D-P-ABORT indication, if the CPDLC-ground-ASE is not in the *IDLE* state, the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the CPDLC-provider-abort service *Reason* parameter set to the abstract value “communication-service-failure”,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.6 CPDLC-ground-ASE Exception Handling

2.3.5.6.1 A Timer Expires

2.3.5.6.1.1 If a CPDLC-ground-ASE detects that a timer has expired, that CPDLC-ground-ASE shall:

- a) Interrupt any current activity,
- b) Create a GroundPDUs APDU with a CPDLCProviderAbortReason [timer-expired] APDU message element,
- c) Invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and
- d) If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the abstract value “timer-expired” as the CPDLC-provider abort service *Reason* parameter value”,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and

- f) Enter the *IDLE* state.

2.3.5.6.2 Unrecoverable System Error

2.3.5.6.2.1 **Recommendation.** — *If a CPDLC-ground-ASE has an unrecoverable system error, the CPDLC-ground-ASE should:*

- a) *Stop any timer,*
- b) *Create a GroundPDUs APDU with a CPDLCProviderAbortReason [undefined-error] APDU message element,*
- c) *Invoke D-ABORT request with:*
 - 1) *the abstract value “provider” as the D-ABORT Originator parameter value, and*
 - 2) *the APDU as the D-ABORT User Data parameter value, and*
- d) *If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the abstract value “undefined-error” as the CPDLC-provider abort service Reason parameter value”,*
- e) *If DSC has the abstract value “true”, set DSC to the abstract value “false”, and*
- f) *Enter the IDLE state.*

2.3.5.6.3 Invalid PDU

2.3.5.6.3.1 If the *User Data* parameter of a D-END confirmation with *Result* parameter set to the abstract value “rejected”, or the *User Data* parameter of a D-START confirmation with *Result* parameter set to the abstract value "accepted", a D-START indication, a D-DATA indication, or a D-END indication, does not contain a valid PDU, the CPDLC-ground-ASE shall:

- a) *Stop any timer,*
- b) *Create a GroundPDUs APDU with a CPDLCProviderAbortReason [invalid-PDU] APDU message element,*
- c) *Invoke D-ABORT request with:*
 - 1) *the abstract value “provider” as the D-ABORT Originator parameter value, and*
 - 2) *the APDU as the D-ABORT User Data parameter value, and*

- d) If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the abstract value “invalid-PDU” as the CPDLC-provider abort service *Reason* parameter value,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.6.3.2 If the *User Data* parameter of a D-START confirmation with *Result* set to the abstract value “rejected (permanent)”, or a D-END confirmation with *Result* set to the abstract value “accepted”, is not a valid PDU the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the CPDLC-provider-abort service *Reason* parameter set to the abstract value “invalid-PDU”,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.6.4 Protocol Error

2.3.5.6.4.1 If the *User Data* parameter of a D-START indication, a D-START confirmation, a D-DATA indication, or a D-END indication is a valid PDU, but is not a PDU for which action is described within a given state as defined in 2.3.5.5, the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) Create a GroundPDUs APDU with a CPDLCProviderAbortReason [protocol-error] APDU message element,
- c) Invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and
- d) If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the abstract value “protocol-error” as the CPDLC-provider abort service *Reason* parameter value”,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.6.4.2 (deleted)

2.3.5.6.4.3 If the *User Data* parameter of a D-END confirmation is a valid PDU, but is not a permitted PDU for which action is described within a given state as defined in 2.3.5.5, the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) If the D-END *Result* parameter is set to the abstract value “rejected”, then
 - 1) Create a GroundPDUs APDU with a CPDLCProviderAbortReason [protocol-error] APDU message element,
 - 2) Invoke D-ABORT request with:
 - iv) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - v) the APDU as the D-ABORT *User Data* parameter value, and
- c) If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the abstract value “protocol-error” as the CPDLC-provider-abort service *Reason* parameter value,
- d) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- e) Enter the *IDLE* state.

2.3.5.6.4.4 Upon receipt of a Dialogue service primitive for which there are no instruction in 2.3.5.3 (i.e. the primitive was not expected or was expected under other conditions or with other parameter values), the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) Create a GroundPDUs APDU with a CPDLCProviderAbortReason [protocol-error] APDU message element,
- c) If a dialogue exists, invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and
- d) If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the abstract value “protocol-error” as the CPDLC-provider-abort service *Reason* parameter value,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and

- f) Enter the *IDLE* state.

2.3.5.6.5 D-START Confirmation Result or Reject Source Not as Expected

2.3.5.6.5.1 If a D-START confirmation *Result* parameter has the abstract value of “rejected (transient)” or if the *Reject Source* parameter has the abstract value of “DS provider”, the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) if the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the CPDLC-provider-abort service *Reason* parameter set to the abstract value “communication-service-error”,
- c) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- d) Enter the *IDLE* state.

2.3.5.6.6 D-START Indication Quality of Service Not as Expected

2.3.5.6.6.1 If a D-START indication *QOS Priority* parameter does not have the abstract value of “high priority flight safety messages” or if the *QOS Residual Error Rate* parameter does not have the abstract value of “low”, or if the *QOS Routing Class* parameter does not have one of the abstract values specified in Table 2.3.6-1, the CPDLC-ground-ASE shall:

- a) Stop any timer,
- b) Create a GroundPDUs APDU with a CPDLCProviderAbortReason [invalid-QOS-parameter] APDU message element,
- c) Invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and
- d) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- e) Enter the *IDLE* state.

2.3.5.6.7 Expected PDU Missing

2.3.5.6.7.1 If the *User Data* parameter of a D-START indication or confirmation, a D-DATA indication, or a D-END indication or confirmation does not contain a PDU, the CPDLC-ground-ASE shall:

- a) Stop any timer,

- b) Create a GroundPDUs APDU with a CPDLCProviderAbortReason [expected-PDU-missing] APDU message element,
- c) Invoke D-ABORT request with:
 - 1) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - 2) the APDU as the D-ABORT *User Data* parameter value, and
- d) If the CPDLC-Ground-User is an active user, invoke CPDLC-provider-abort service indication with the abstract value “not-permitted-PDU” as the CPDLC-provider-abort service *Reason* parameter value,
- e) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and
- f) Enter the *IDLE* state.

2.3.5.6.8 D-START Security Requirements Parameter Not as Expected

Note.— This section applies to the case when the D-START Security Requirements parameter does not meet the local security policy (as defined in Annex 10 Volume III Part I Chapter 3).

2.3.5.6.8.1 Upon receipt of a D-START indication with the *Security Requirements* parameter not consistent with the local security policy (as defined in Annex 10 Volume III Part I Chapter 3) or upon receipt of a D-START confirmation with the *Security Requirements* parameter not equal to the value that was set in the D-START request, the CPDLC-ground-ASE shall::

- a) Stop all timers,
- b) If the dialogue with the peer is still open:
 - 1) Create a GroundPDUs APDU with a CPDLCProviderAbortReason [communication-service-failure] APDU message element,
 - 2) Invoke D-ABORT request with:
 - i) the abstract value “provider” as the D-ABORT *Originator* parameter value, and
 - ii) the APDU as the D-ABORT *User Data* parameter value,
- c) If the CPDLC-ground-user is active, invoke CPDLC-provider-abort service indication with the abstract value “communication-service-failure” as the CDPLC-provider-abort service *Reason* parameter value,
- d) If DSC has the abstract value “true”, set DSC to the abstract value “false”, and

- e) Enter the *IDLE* state.

2.3.5.7 CPDLC ASE State Tables

2.3.5.7.1 Priority

2.3.5.7.1.1 If the state tables shown for the CPDLC-air-ASE and the CPDLC-ground-ASE shown below conflict with textual statements made elsewhere in this document, the textual statements shall take precedence.

Note 1.— In the following state tables, the statement “cannot occur” means that if the implementation conforms to the SARPs, it is impossible for this event to occur. If the event does occur, this implies that there is an error in the implementation. If such a situation is detected, it is suggested that the ASE aborts with the error “unrecoverable system error”.

Note 2.— In the following state tables, the statement “not permitted” means that the implementation must prevent this event from occurring through some local means. If the event does occur this implies that there is an error in the implementation. If such a situation is detected, it is suggested that the ASE performs a local rejection of the request rather than aborting the dialogue.

Table 2.3.5-2. CPDLC-air-ASE State Table

STATE → EVENT ↓	IDLE	START-REQ	START-IND	DIALOGUE	END
Dialogue Service Events					
D-START Indication APDU = ProtectedUplinkMessage (startup)	● CPDLC-start indication ⇒ <i>START-IND</i>	cannot occur	cannot occur	cannot occur	cannot occur
D-START Confirmation <i>Result</i> “accepted”, DSC = “false” APDU = ProtectedUplinkMessage (send)	cannot occur	● Stop timer t_{start} ● CPDLC-start confirmation ⇒ <i>DIALOGUE</i>	cannot occur	cannot occur	cannot occur
D-START Confirmation <i>Result</i> “rejected (permanent)” and Reject Source “DS user”, DSC=“false” APDU = ProtectedUplinkMessage (send)	cannot occur	● Stop timer t_{start} ● CPDLC-start confirmation ⇒ <i>IDLE</i>	cannot occur	cannot occur	cannot occur
D-START Confirmation <i>Result</i> “accepted”, DSC = “true” APDU = ProtectedUplinkMessage (send)	cannot occur	● Stop timer t_{start} ● DSC-start confirmation, ⇒ <i>DIALOGUE</i>	cannot occur	cannot occur	cannot occur
D-START Confirmation <i>Result</i> “rejected (permanent)” and Reject Source “DS user”, DSC= “true” APDU = ProtectedUplinkMessage (send)	cannot occur	● Stop timer t_{start} ● DSC-start confirmation ● Set DSC = “false” ⇒ <i>IDLE</i>	cannot occur	cannot occur	cannot occur
D-DATA Indication APDU = ProtectedUplinkMessage (send)	cannot occur	cannot occur	cannot occur	● CPDLC-message indication ⇒ <i>DIALOGUE</i>	● if DSC=“true” ● CPDLC-message indication ⇒ <i>END</i> ● else cannot occur
D-END Indication: DSC=“false” APDU = ProtectedUplinkMessage (send)	cannot occur	cannot occur	cannot occur	● CPDLC-end indication ⇒ <i>END</i>	cannot occur

STATE → EVENT ↓	IDLE	START-REQ	START-IND	DIALOGUE	END
D-END Confirmation: DSC="true" Result "accepted" APDU = ProtectedUplinkMessage (send)	cannot occur	cannot occur	cannot occur	cannot occur	● DSC-end confirmation ● Set DSC "false" ⇒ IDLE
D-END Confirmation: DSC="false", Result "rejected" APDU = ProtectedUplinkMessage (send)	cannot occur	cannot occur	cannot occur	cannot occur	● DSC-end confirmation ⇒ DIALOGUE
CPDLC-User Events					
CPDLC-start Request	● D-START request ● Start timer t_{start} ⇒ START-REQ	not permitted	not permitted	not permitted	not permitted
CPDLC-start Response DSC="false" Result "accepted"	not permitted	not permitted	● D-START response ⇒ DIALOGUE	not permitted	not permitted
CPDLC-start Response DSC="false" Result "rejected"	not permitted	not permitted	● D-START response ⇒ IDLE	not permitted	not permitted
DSC-start Request	● D-START request ● set DSC "true" ● Start timer t_{start} ⇒ START-REQ	not permitted	not permitted	not permitted	not permitted
CPDLC-message Request	not permitted	not permitted	not permitted	● D-DATA request ⇒ DIALOGUE	● if DSC="false" ● D-DATA request ⇒ END ● else not permitted
CPDLC-end Service Response DSC = "false" Result "accepted"	cannot occur	cannot occur	cannot occur	not permitted	● D-END response ⇒ IDLE
CPDLC-end Service Response DSC = "false" Result "rejected"	cannot occur	cannot occur	cannot occur	not permitted	● D-END response ⇒ DIALOGUE

STATE → EVENT ↓	IDLE	START-REQ	START-IND	DIALOGUE	END
DSC-end Request: DSC = “true”	not permitted	not permitted	not permitted	●D-END request ⇒END	not permitted
ABORT Events					
CPDLC-user-abort Request	not permitted	●Stop timer t_{start} if set ●D-ABORT request ●If DSC = “true”, set DSC = “false” ⇒IDLE	●Stop timer t_{start} if set ●D-ABORT request ●If DSC = “true”, set DSC = “false” ⇒IDLE	●D-ABORT request ●If DSC = “true”, set DSC = “false” ⇒IDLE	●D-ABORT request ●If DSC = “true”, set DSC = “false” ⇒IDLE
D-ABORT Indication <i>Originator</i> “user”	cannot occur	●Stop timer T_{start} if set ●If active user: CPDLC-user-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	●Stop timer T_{start} if set ●If active user: CPDLC-user-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	●If active user: CPDLC-user-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	●If active user: CPDLC-user-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE
D-ABORT Indication <i>Originator</i> “provider”	cannot occur	●Stop timer T_{start} if set ●If active user: CPDLC-provider-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	●Stop timer T_{start} if set ●If active user: CPDLC-provider-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	●If active user: CPDLC-provider-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	●If active user: CPDLC-provider-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE
D-P-ABORT indication	cannot occur	●Stop timer t_{start} if set ●If active user: CPDLC-provider-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	●Stop timer t_{start} if set ●If active user: CPDLC-provider-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	●If active user: CPDLC-provider-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	●If active user: CPDLC-provider-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE
T_{start} Expires	cannot occur	●D-ABORT request ●CPDLC-provider-abort indication ●If DSC = “true”, set DSC = “false” ⇒IDLE	cannot occur	cannot occur	cannot occur

Table 2.3.5-3. CPDLC-ground-ASE State Table

STATE → EVENT ↓	IDLE	START-REQ	START-IND	DIALOGUE	END	FORWARD
Dialogue Service Events						
D-START Indication APDU Aircraft mode “cpdlc”	● CPDLC-start indication ⇒ <i>START-IND</i>	cannot occur	cannot occur	cannot occur	cannot occur	cannot occur
D-START Indication APDU Aircraft mode “dsc”	● DSC-start indication, ● Set DSC = “true” ⇒ <i>START-IND</i>	cannot occur	cannot occur	cannot occur	cannot occur	cannot occur
D-START Indication APDU Forward version equal and function supported	● CPDLC-forward indication ● D-START response ⇒ <i>IDLE</i>	cannot occur	cannot occur	cannot occur	cannot occur	cannot occur
D-START Indication APDU Forward version not equal or function not supported	● D-START response ⇒ <i>IDLE</i>	cannot occur	cannot occur	cannot occur	cannot occur	cannot occur
D-START Confirmation <i>Result</i> “accepted” DSC = “false”	cannot occur	● Stop timer t_{start} ● CPDLC-start confirmation ⇒ <i>DIALOGUE</i>	cannot occur	cannot occur	cannot occur	cannot occur
D-START Confirmation <i>Result</i> “rejected (permanent)” and <i>Reject Source</i> “DS user” DSC = “false”	cannot occur	● Stop timer t_{start} ● CPDLC-start confirmation ⇒ <i>IDLE</i>	cannot occur	cannot occur	cannot occur	● Stop timer t_{start} ● CPDLC-forward confirmation ⇒ <i>IDLE</i>
D-DATA Indication	cannot occur	cannot occur	cannot occur	● CPDLC-message indication ⇒ <i>DIALOGUE</i>	● if DSC = “false” ● CPDLC-message indication ⇒ <i>END</i> ● else not permitted	cannot occur

STATE → EVENT ↓	IDLE	START-REQ	START-IND	DIALOGUE	END	FORWARD
D-END Indication: DSC="true"	cannot occur	cannot occur	cannot occur	●DSC-end indication ⇒END	cannot occur	cannot occur
D-END Confirmation: DSC = "false" Result "accepted"	cannot occur	cannot occur	cannot occur	cannot occur	●CPDLC-end confirmation ⇒IDLE	cannot occur
D-END Confirmation: DSC = "false" Result "rejected"	cannot occur	cannot occur	cannot occur	cannot occur	●CPDLC-end confirmation ⇒DIALOGUE	cannot occur
CPDLC-User Events						
CPDLC-start Request	●D-START request ●Start timer t_{start} ⇒START-REQ	not permitted	not permitted	not permitted	not permitted	not permitted
CPDLC-start Response DSC = "false" Result "accepted"	not permitted	not permitted	●D-START response ⇒DIALOGUE	not permitted	not permitted	not permitted
CPDLC-start Response DSC = "false" Result "rejected"	not permitted	not permitted	●D-START response ⇒IDLE	not permitted	not permitted	not permitted
DSC-start Response DSC = "true" Result "accepted"	not permitted	not permitted	●D-START response ⇒DIALOGUE	not permitted	not permitted	not permitted
DSC-start Response DSC = "true" Result "rejected"	not permitted	not permitted	●D-START response ●Set DSC= "false" ⇒IDLE	not permitted	not permitted	not permitted
CPDLC-message Request	not permitted	not permitted	not permitted	●D-DATA request ⇒DIALOGUE	●If DSC= "true" ●D-DATA request ⇒END ●Else not permitted	not permitted

STATE → EVENT ↓	IDLE	START-REQ	START-IND	DIALOGUE	END	FORWARD
CPDLC-end Request: DSC = “false”	not permitted	not permitted	not permitted	●D-END request ⇒END	not permitted	not permitted
DSC-end Service Response DSC = “true” Result “accepted”	cannot occur	cannot occur	cannot occur	not permitted	●D-END response ●Set DSC = “false” ⇒IDLE	not permitted
DSC-end Service Response DSC = “true” Result “rejected”	cannot occur	cannot occur	cannot occur	not permitted	●D-END response ⇒DIALOGUE	not permitted
CPDLC-forward Request	●D-START request ●Start timer t_{start} ⇒FORWARD	not permitted	not permitted	not permitted	not permitted	not permitted
ABORT Events						
CPDLC-user-abort Request	not permitted	●Stop timer t_{start} , if set ●D-ABORT request ●If DSC = “true”, set DSC= “false” ⇒IDLE	●Stop timer t_{start} , if set ●D-ABORT request ●If DSC = “true”, set DSC= “false” ⇒IDLE	●D-ABORT request ●If DSC = “true”, set DSC= “false” ⇒IDLE	●D-ABORT request ●If DSC = “true”, set DSC= “false” ⇒IDLE	●Stop timer t_{start} , if set ●D-ABORT request ⇒IDLE
D-ABORT Indication <i>Originator</i> “provider”	cannot occur	●Stop timer T_{start} , if set ●If active user: CPDLC-provide r-abort indication ●If DSC = “true”, set DSC= “false” ⇒IDLE	●Stop timer T_{start} , if set ●If active user: CPDLC-provide r-abort indication ●If DSC = “true”, set DSC= “false” ⇒IDLE	●If active user: CPDLC-provide r-abort indication ●If DSC = “true”, set DSC= “false” ⇒IDLE	●If active user: CPDLC-provide r-abort indication ●If DSC = “true”, set DSC= “false” ⇒IDLE	●Stop timer T_{start} , if set ●If active user: CPDLC-provide r-abort indication ⇒IDLE
D-ABORT Indication <i>Originator</i> “user”	cannot occur	●Stop timer T_{start} , if set ●If active user: CPDLC-user-abort indication ●If DSC = “true”, set DSC= “false” ⇒IDLE	●Stop timer T_{start} , if set ●If active user: CPDLC-user-abort indication ●If DSC = “true”, set DSC= “false” ⇒IDLE	●If active user: CPDLC-user-abort indication ●If DSC = “true”, set DSC= “false” ⇒IDLE	●If active user: CPDLC-user-abort indication ●If DSC = “true”, set DSC= “false” ⇒IDLE	●Stop timer T_{start} , if set ●If active user: CPDLC-user-abort indication ⇒IDLE

STATE → EVENT ↓	IDLE	START-REQ	START-IND	DIALOGUE	END	FORWARD
D-P-ABORT indication	cannot occur	● Stop timer t_{start}, if set ● If active user: CPDLC-provide r-abort indication ● If DSC = "true", set DSC= "false" ⇒ IDLE	● Stop timer t_{start}, if set ● If active user: CPDLC-provide r-abort indication ● If DSC = "true", set DSC= "false" ⇒ IDLE	● If active user: CPDLC-provide r-abort indication ● If DSC = "true", set DSC= "false" ⇒ IDLE	● If active user: CPDLC-provide r-abort indication ● If DSC = "true", set DSC= "false" ⇒ IDLE	● Stop timer t_{start}, if set ● If active user: CPDLC-provide r-abort indication ⇒ IDLE
T_{start} Expires	cannot occur	● D-ABORT request ● CPDLC-provide r-abort indication ● If DSC = "true", set DSC= "false" ⇒ IDLE	cannot occur	cannot occur	cannot occur	● D-ABORT request ● CPDLC-provide r-abort indication ⇒ IDLE